Virtuoso® Layout Pro: T2 Create and Edit Commands

Course Version IC 25.1

Lab Manual **Revision 1.0**

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Virtuoso Layout Pro: T2 Create and Edit Commands

Module 1: About This Course

There are no labs for this module.

Module 2: Create and Edit Commands

Lab 2-1 Using the Create Via Command

Objective: To use the advanced via creation features.

In this lab, you use the *Auto Via* feature to quickly create vias for your layout.

Lab Setup

Follow the steps below to set up the environment and perform the lab:

1. Untar the lab database file using the following command.

   ```
   tar -xzvf <filename.tar.gz>
   ```

2. Update the following line in the /VLPT2_IC_25_1/. cshrc_VLPT2 file with the appropriate valid local tool paths for Virtuoso®.

   ```
   setenv CDSHOME /ccstools/cdsind1/Software/IC251Base_lnx86
   ```

3. Save and close the file.

4. Source the updated class setup file.

   ```
   source .cshrc_VLPT2
   ```

Invoking the *virtuoso* Executable

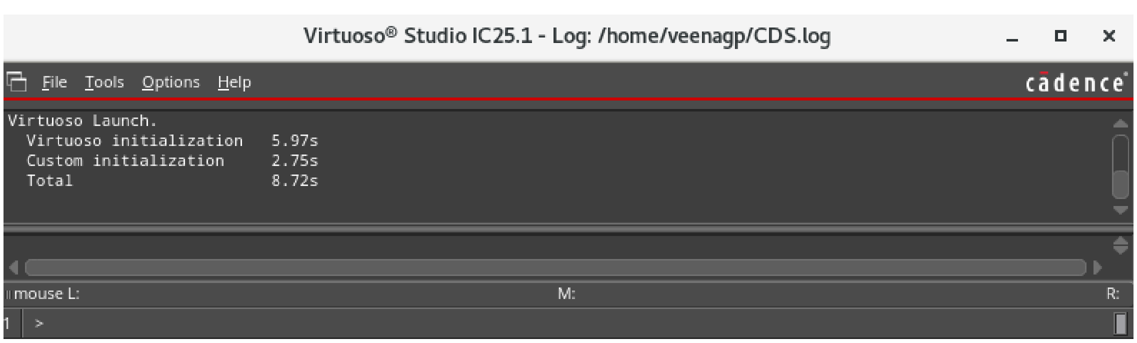
1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

   ```
   cd ./M02_VLPT2_CreateVia
   ```

2. To start the Cadence Virtuoso® software, enter this command in the same terminal window:

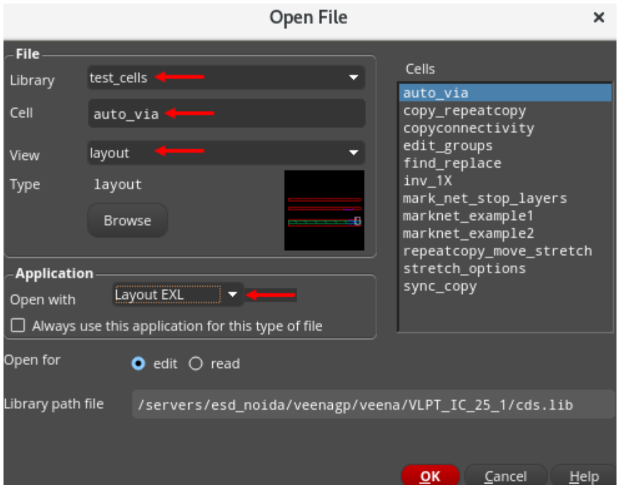
   ```
   virtuoso &
   ```

   The Command Interpreter Window (CIW) appears:



3. In the CIW, choose **File – Open**. Enter the following values in the form.

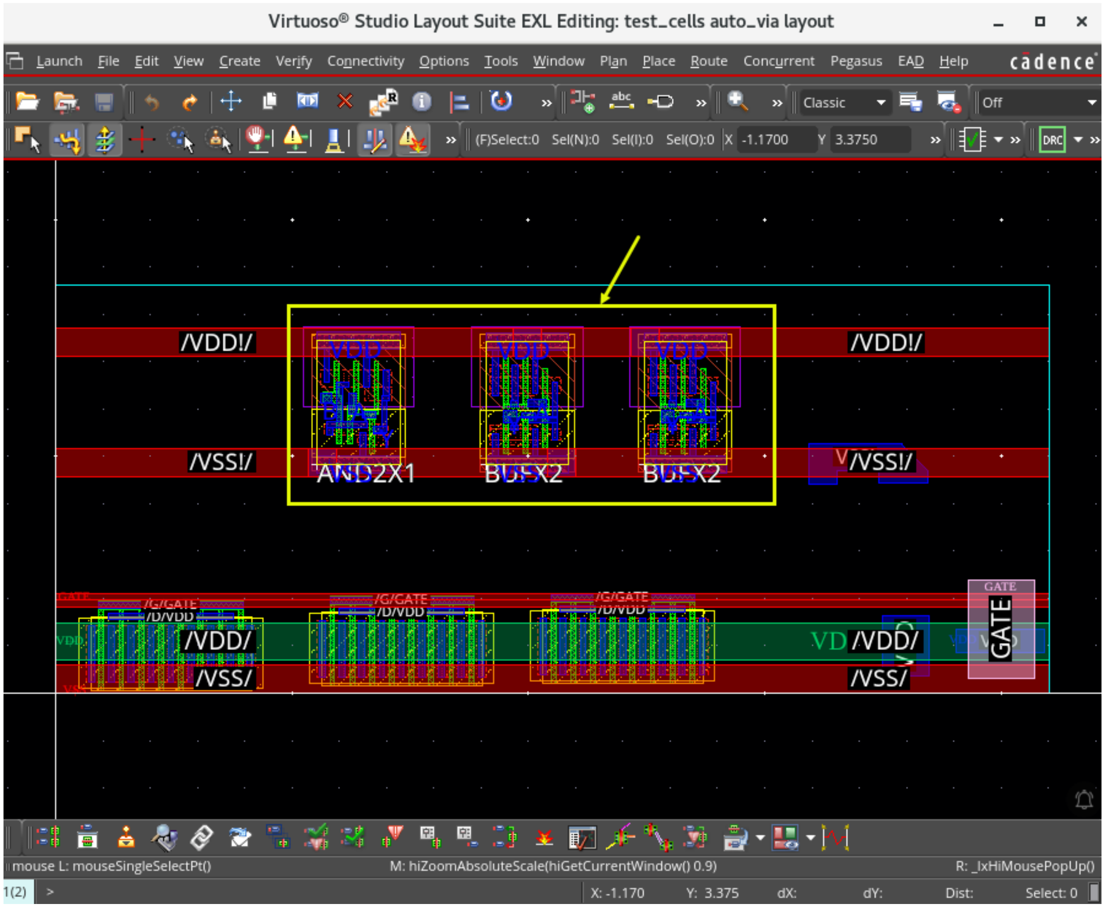
Library	`test_cells`
Cell	`auto_via`
View	`layout`



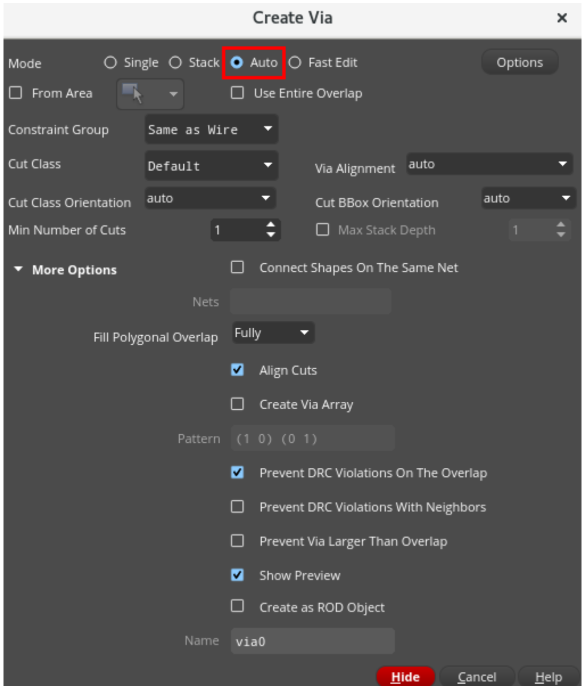
4. Click **OK**.

   The *test_cells/auto_via/schematic* and *test_cells/auto_via/layout* cellviews open in EXL mode. Minimize the schematic view and maximize the layout view.

5. Zoom in to the upper part of the cell, where three standard cells are placed, as shown here.



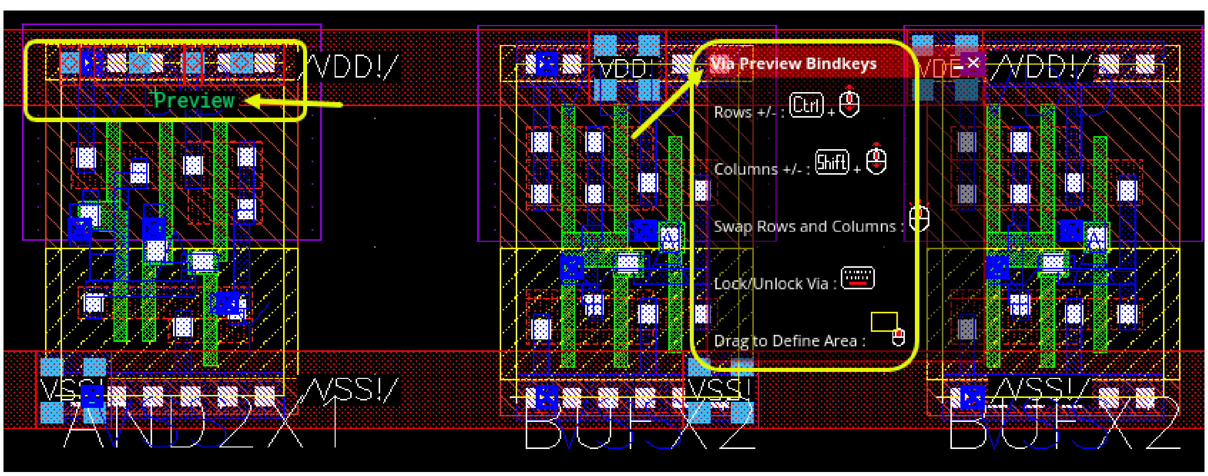
6. Choose **Create – Via** or press the bindkey **O** and select **Auto** in the Create Via form.



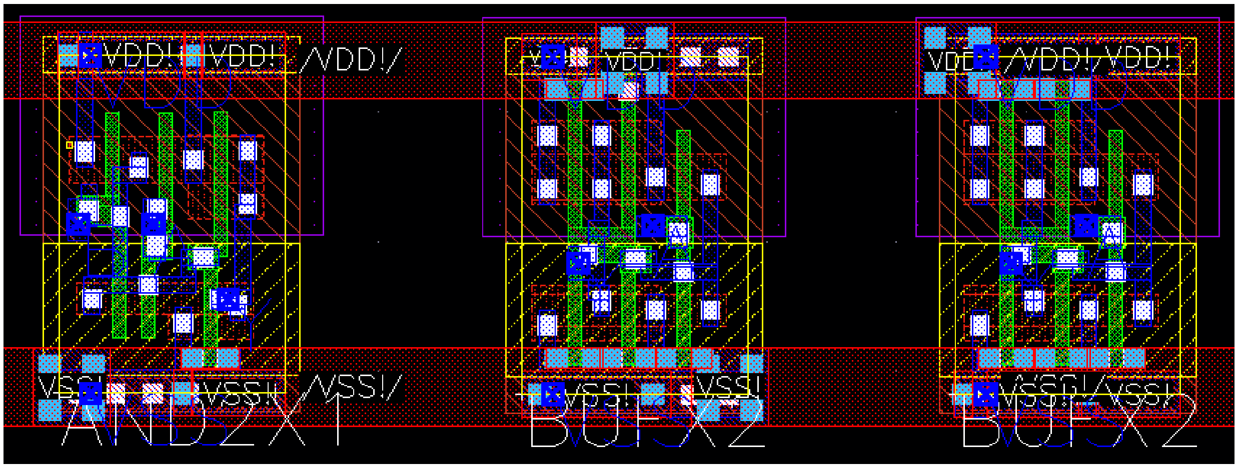
Note: If the Create Via form is not visible, press **F3** to display the form.

7. Keep the Create Via form with the default option and click on the power/ground rails to create the power/ground contacts.

   Using the **Via Preview Bindkeys** form, you can view the **Preview** of the vias placed.



You will get the contacts in the layout, something like what is shown here.

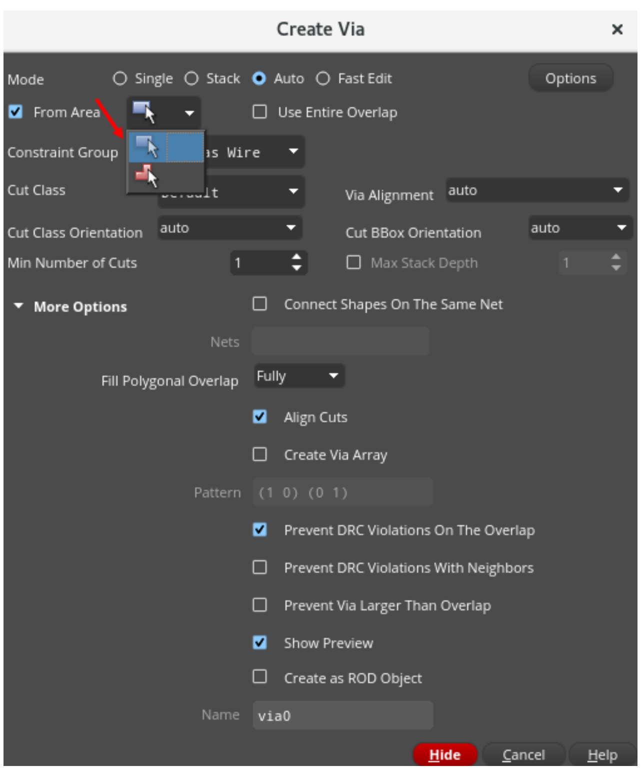


Important: You will notice that in some cases, contact is already there, but when clicking next to it, a new contact will be created. This feature can be used to improve the quality of already existing via regions.

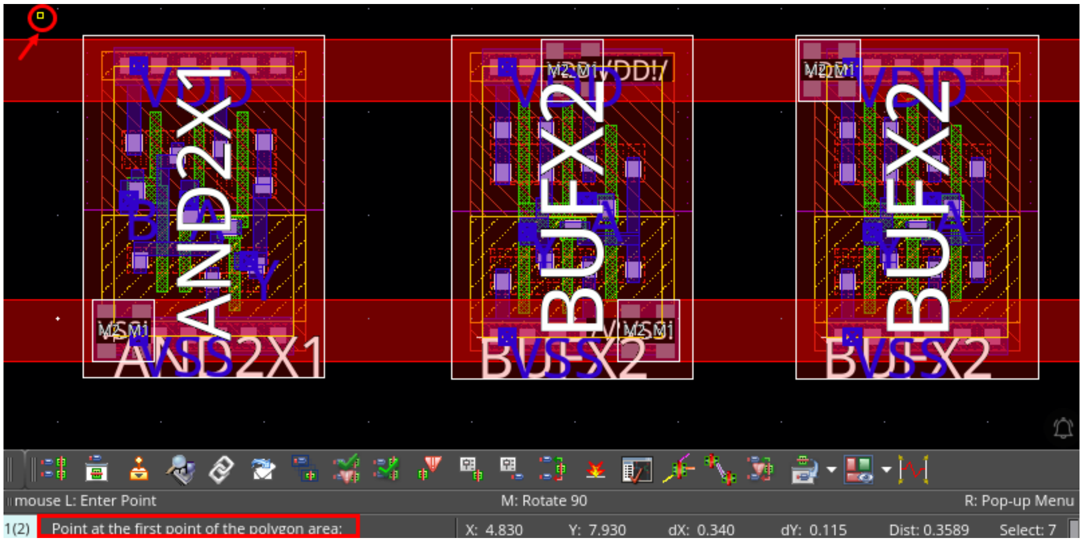
8. Cancel the Create Via form and choose **File – Discard Edits** to discard all edits.

9. Choose **Create – Via** or press the bindkey **O** to invoke the Create Via form and select **Auto** in the Create Via form.

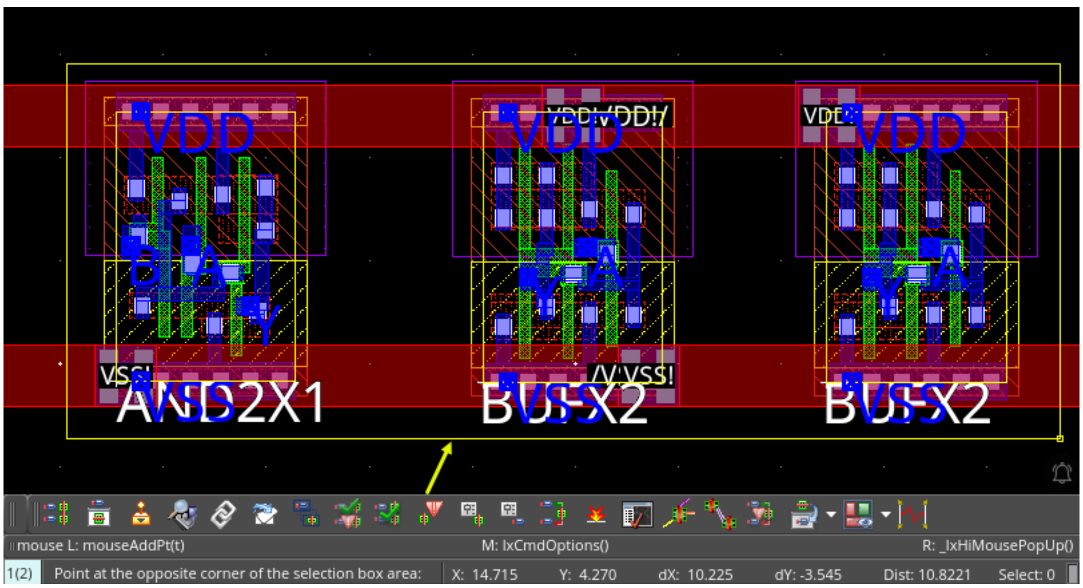
10. Click **From Area** cyclic field in the Create Via form and select **Rectangle**, as shown here.



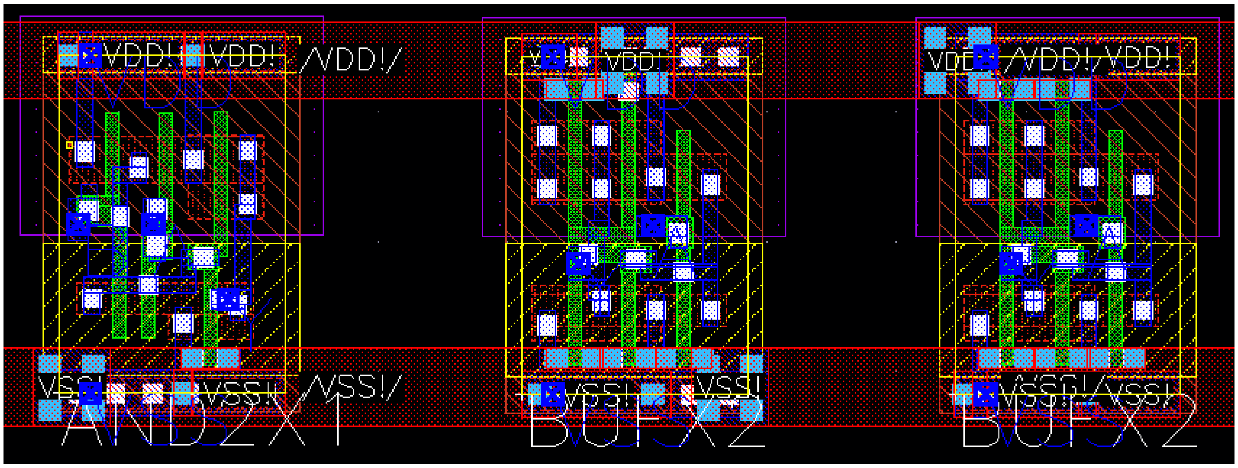
11. **Point at the first corner of the rectangle area**, as shown here.



12. **Point at the opposite corner of the rectangle area**, as shown here.

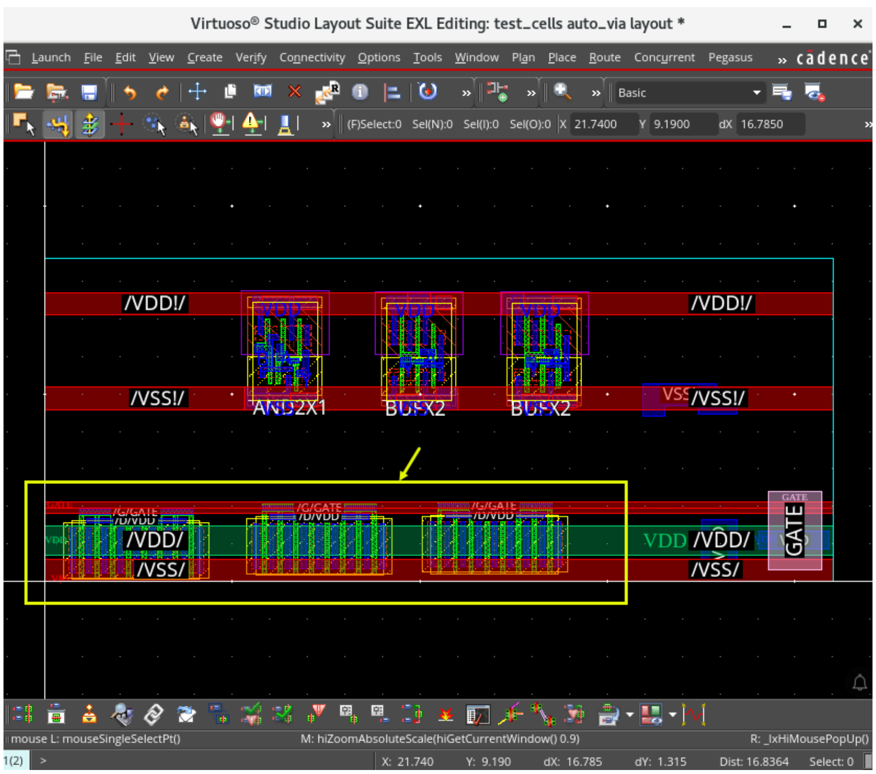


13. See the contacts created, as shown here.

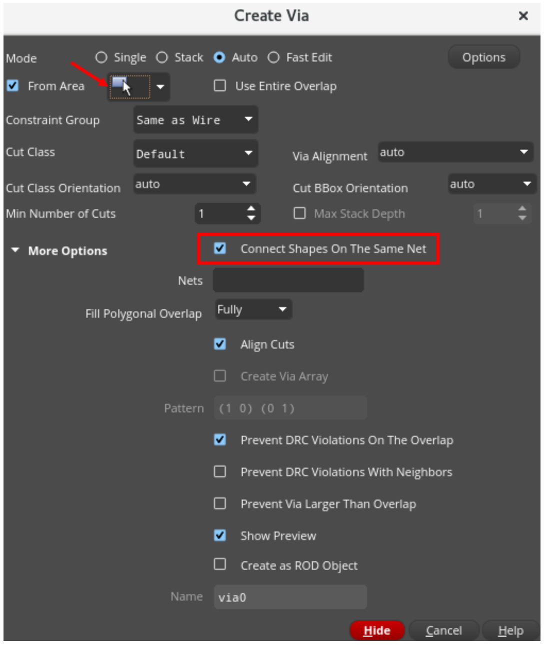


14. Keep the Create Via form open.

15. Zoom in to the bottom area where three transistors are placed, as shown here.

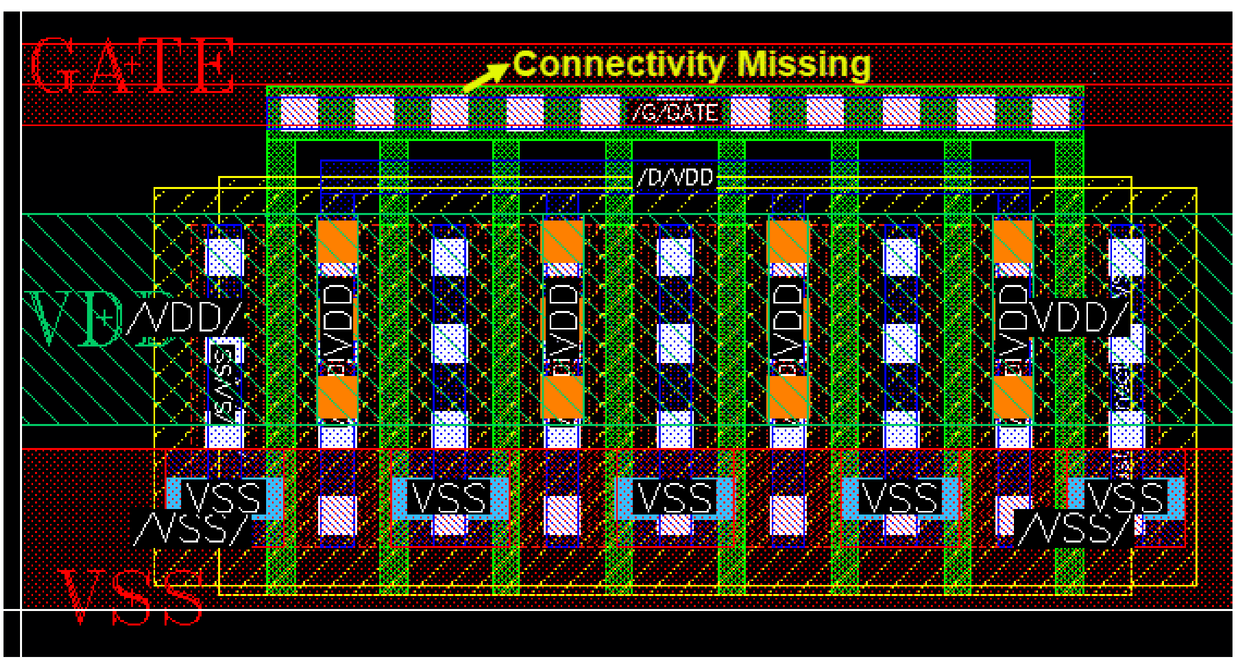


16. Enable the **Connect Shapes On The Same Net** checkbox in the Create Via form, as shown here.

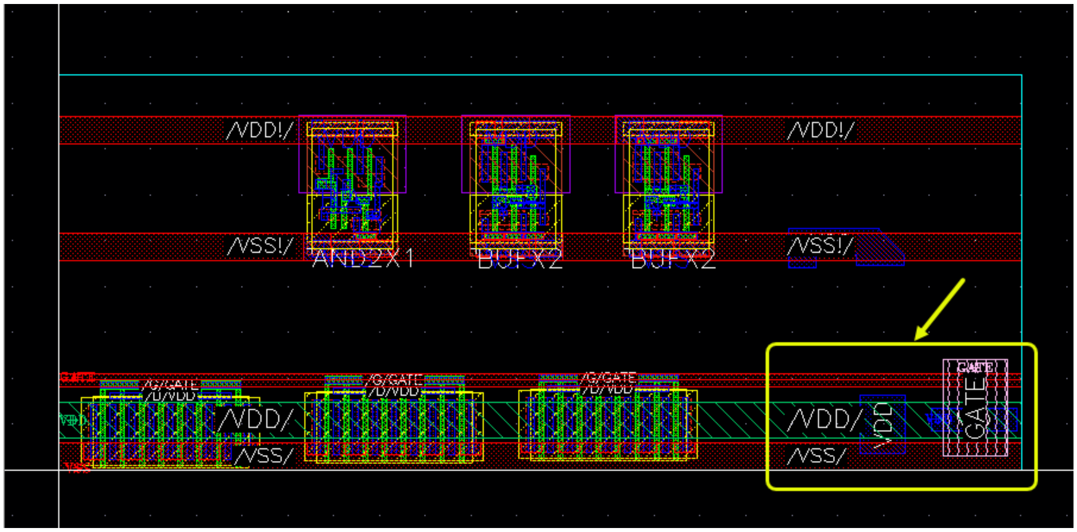


17. Area-select the three transistors as you did in **Step 11** and **Step 12**. You will notice that the Create Via command will only create a contact in the “right” position, avoiding the creation of shorts.

18. If you try now to create the “gate” contacts, the Auto Contact function will not work: this is because the **Metal1** line marked as “GATE” in reality has no connectivity information at all. If the **Connect Shapes On The Same Net** option is active, the vias will be created only between objects which have the same connectivity information.

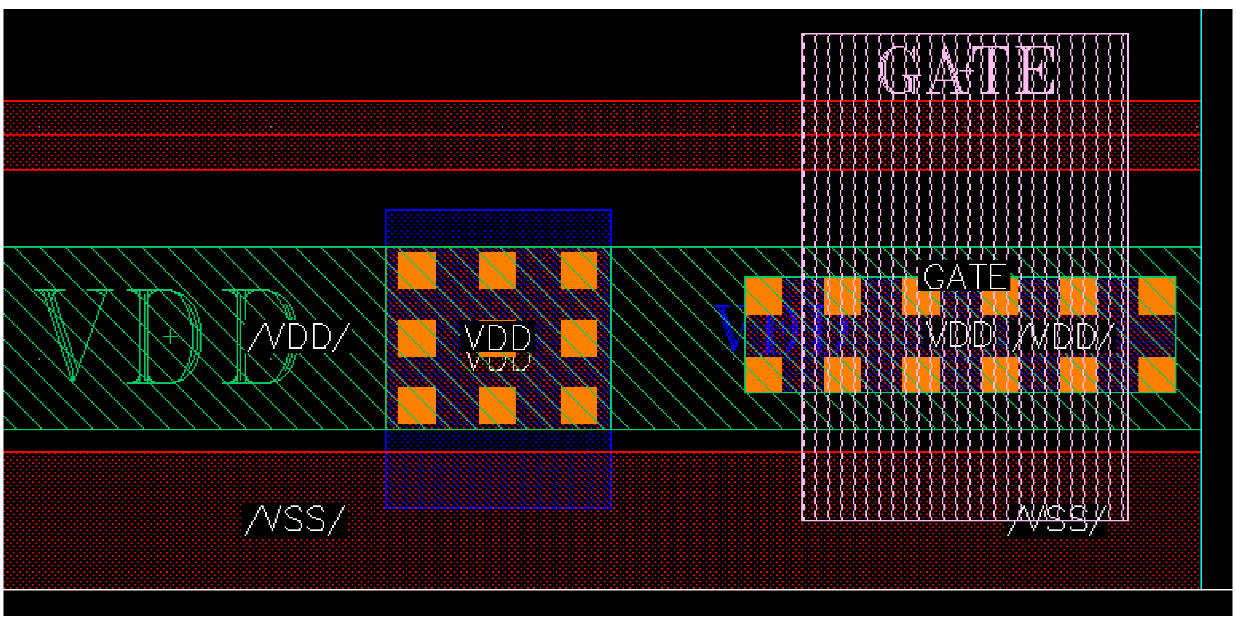


19. Zoom in to the right side of the layout, where you will create stacked vias.



20. Reopen the Create Via form if it is not already open and try creating stacked vias, as you did in **Step 17**.

    The graphic of the stacked vias is shown here.



21. In the Create Via form, in the *From Area* cyclic field, try selecting **Polygon** and explore the via creation as time permits.

22. Close the design window. **Do NOT** save.

23. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you learned how to

- Use the *Auto Via* feature to quickly create vias for your layout

End of Lab

Lab 2-2 Using the Create Label Command

Objective: To use the automatic label creation feature.

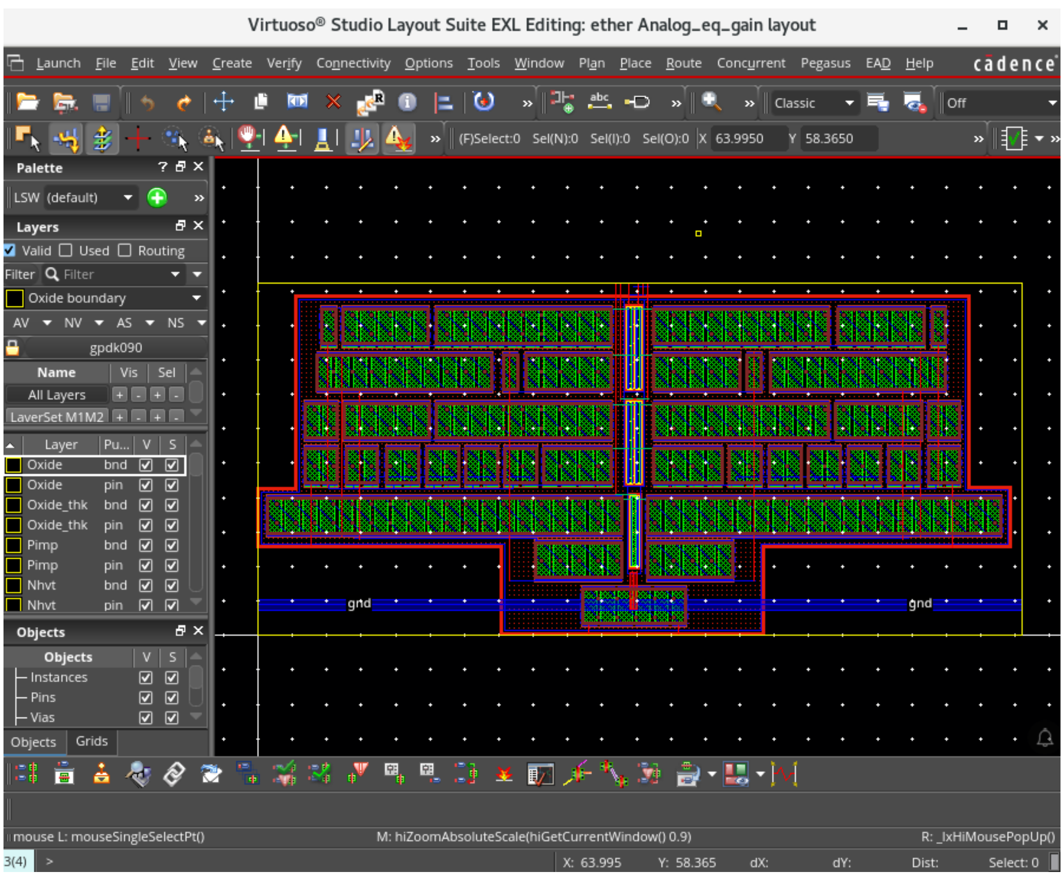
In this lab, you use the *Auto Label* feature to quickly generate labels as *Text Display* and *Label* for your layout.

1. In the CIW, choose **File – Open** and enter the following values in the form.

Library	`ether`
Cell	`Analog_eq_gain`
View	`layout`

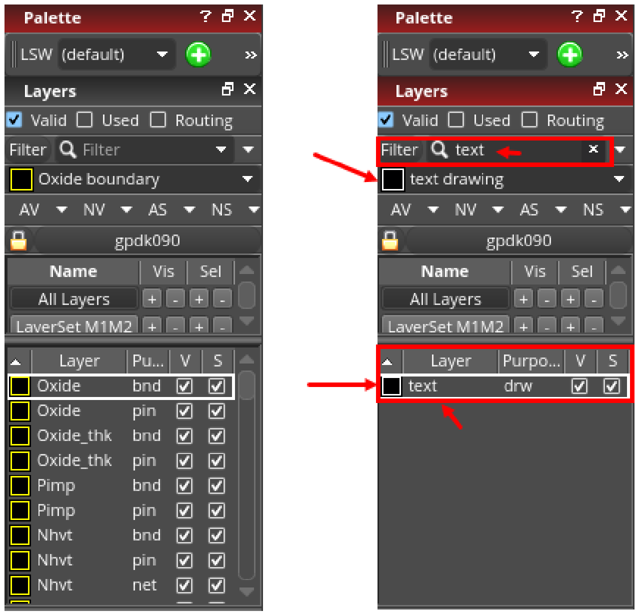
2. Click **OK**.

   The *ether/Analog_eq_gain/schematic* and *ether/Analog_eq_gain/layout* cellviews open in EXL mode. Minimize the schematic view and maximize the layout view.



3. Select the layer **text drawing** in the Layer Palette.

   Note: To find it quickly, just enter **text** in the *Filter* field.



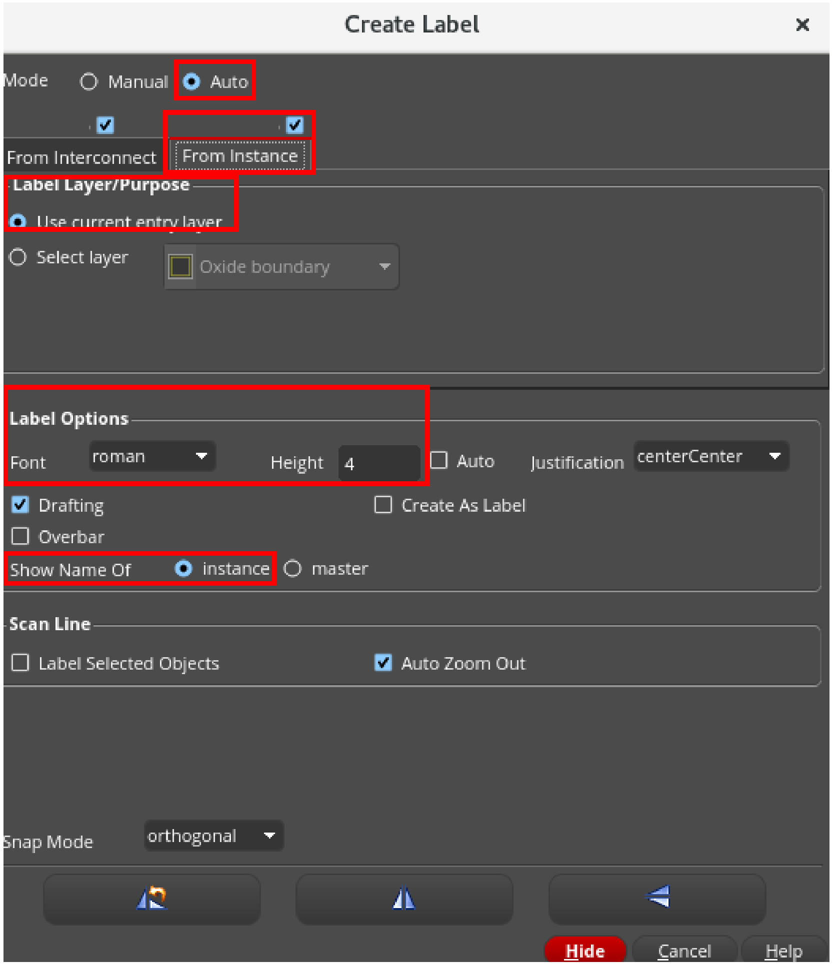
Using the *Auto Label* Feature to Quickly Generate Labels as *Text Display*

1. Choose **Create – Label** or press the bindkey **L** to open the Create Label form.

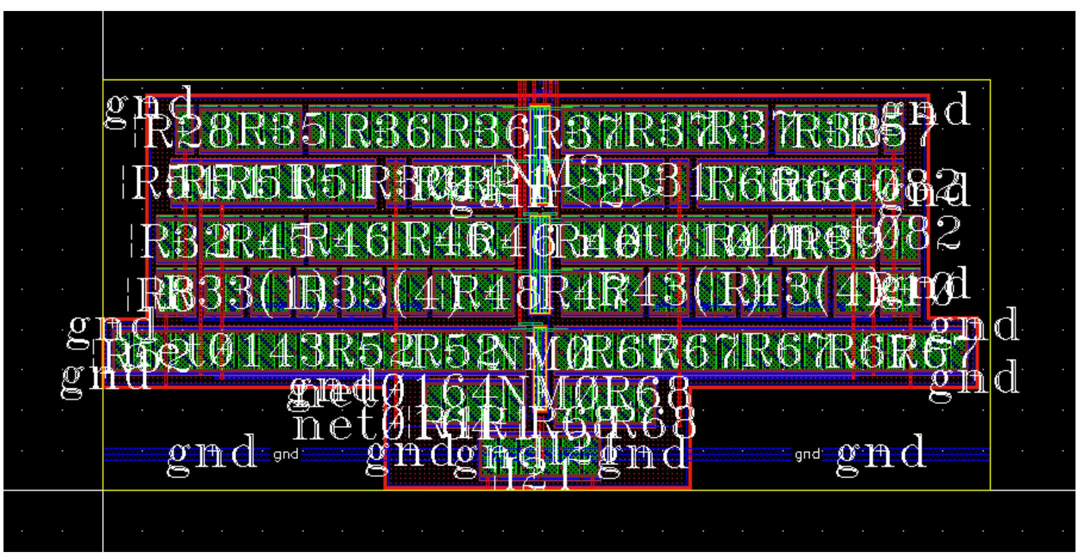
   Note: If the Create Label form is not visible, press **F3** to display the form.

2. In the Create Label form, select the options, as shown here.
   - Mode: **Auto**
   - Tab: **From Instance**
   - Label Layer/Purpose: **Use current entry layer**
   - Font: **roman**
   - Height: **4**
   - Show Name Of: **instance**

3. Hide the Create Label form and click on the instances in the design to label them.



4. In the same way, click on the nets to label them as well.



Important: This command uses the connectivity information generated using a connectivity-driven layout flow. In this flow, once the layout is “clean”, it is also possible to automatically update the name of the layout instances to match the schematic ones.

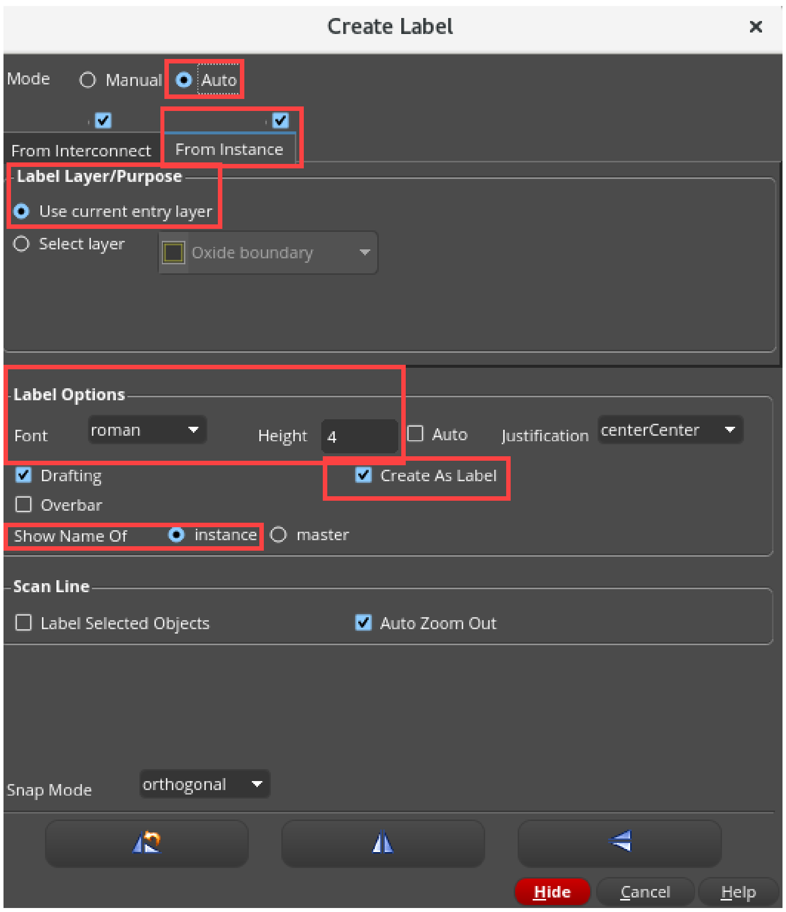
Note: You can check that the properties of the created labels are of type *Text Display* by using the **Edit – Basic – Properties** (bindkey **Q**) command or by using the Property Editor assistant.

5. Click **Cancel** on the Create Label form to cancel the Create Label form.

6. Choose **File – Discard Edits** to discard all the edits.

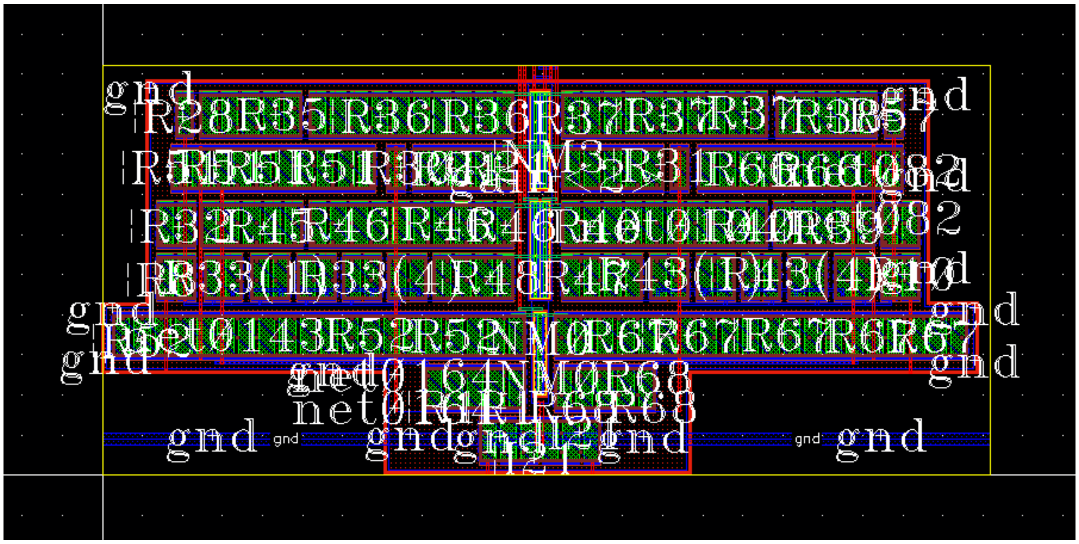
Using the *Auto Label* Feature to Quickly Generate Labels as *Label*

1. Choose **Create – Label** or press the bindkey **L** to open the Create Label form and select the option **Create As Label**, as shown here.



2. Hide the Create Label form and click on the instances in the design to label them.

3. In the same way, click on the nets to label them as well.



Note: You can check that the properties of the created labels are of type *Label* by using the **Edit – Basic – Properties (**bindkey **Q**) command or by using the Property Editor assistant.

4. Click **Cancel** on the Create Label form.

5. Close the design window. **Do NOT save**.

6. Leave the Virtuoso session open for the next lab.

Summary

In this lab, you learned how to

- Use the *Auto Label* feature to quickly generate labels as *Text Display* and *Label* for your layout

End of Lab

Lab 2-3 Using the Copy and Repeat Copy Commands

Objective: To copy and repeat copy objects at a precise dimension from one another.

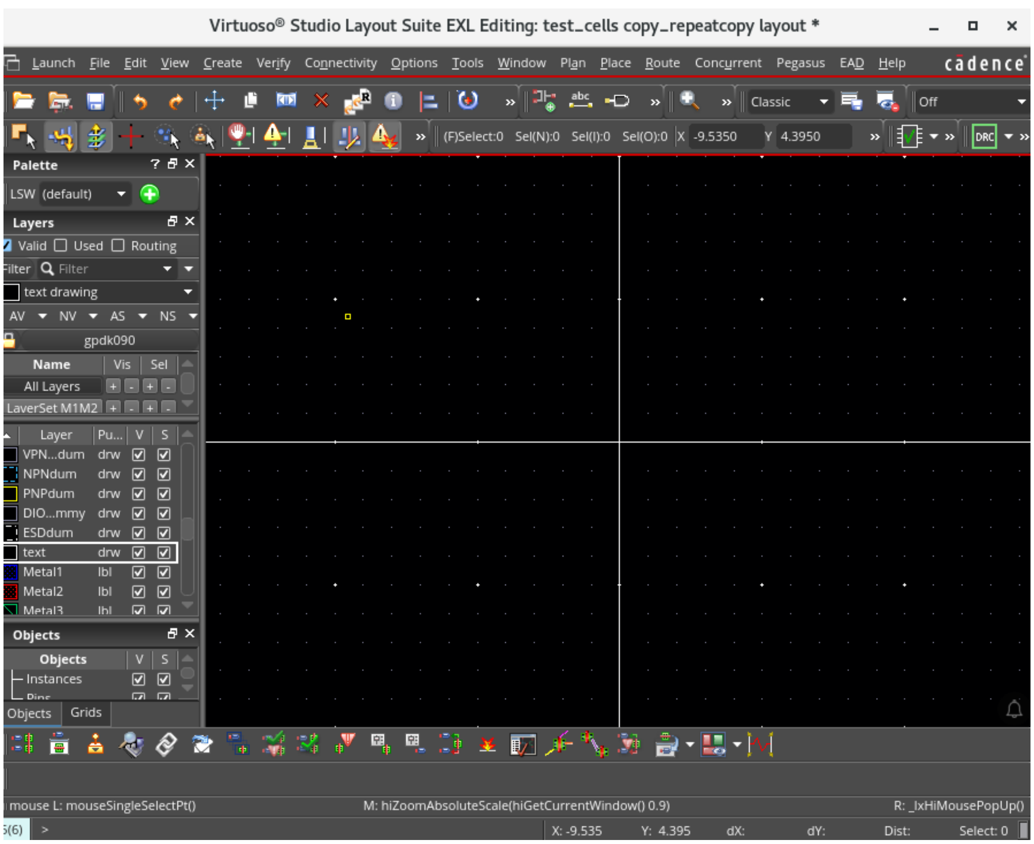
In this lab, you use the *Copy* and *Repeat Copy* commands to copy objects at a precise dimension from one another and use the *Copy connectivity* option in the *Copy* command.

1. In the CIW, choose **File – Open** and enter the following values in the form.

Library	`test_cells`
Cell	`copy_repeatcopy`
View	`layout`

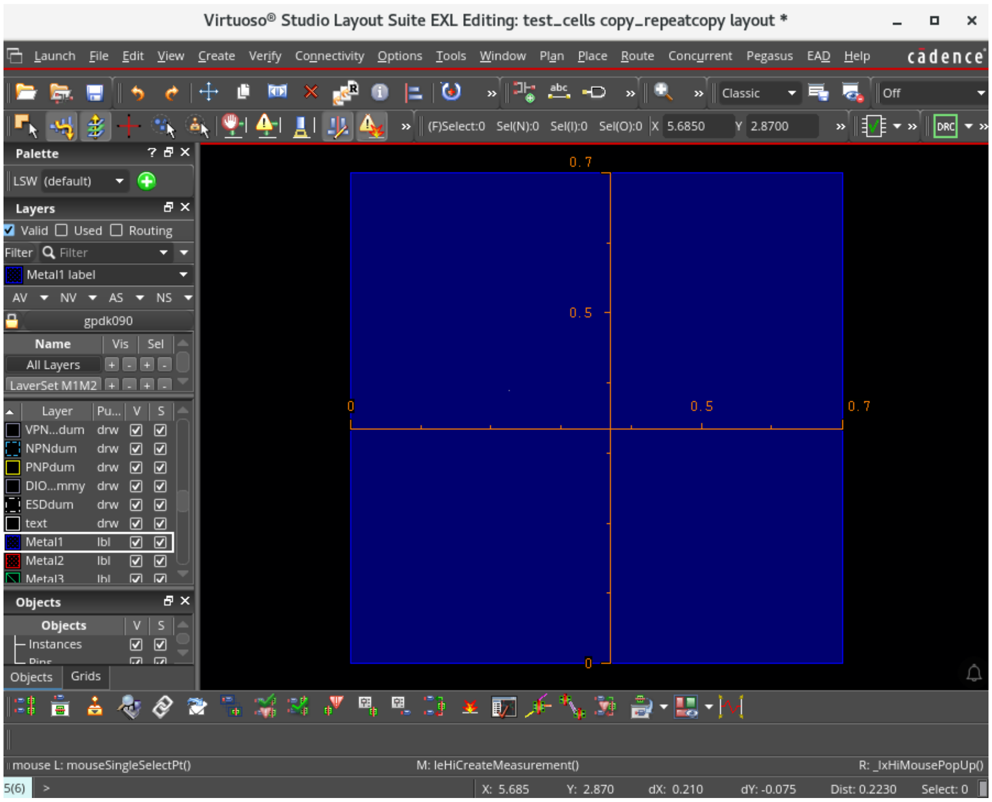
2. Click **OK**.

   The *test_cells/copy_repeatcopy/layout* cellview opens in EXL mode for editing.



3. Choose **Create – Shape – Rectangle** or press the bindkey **R** and create a square on **Metal1** drawing.

   Important: Make the dimensions 0.7 x 0.7, as shown here.



Note: You can use the **Tools – Create Measurement** command or the bindkey **K** to measure the square, and the bindkey **Shift+K** to delete the Ruler when you are through with it.

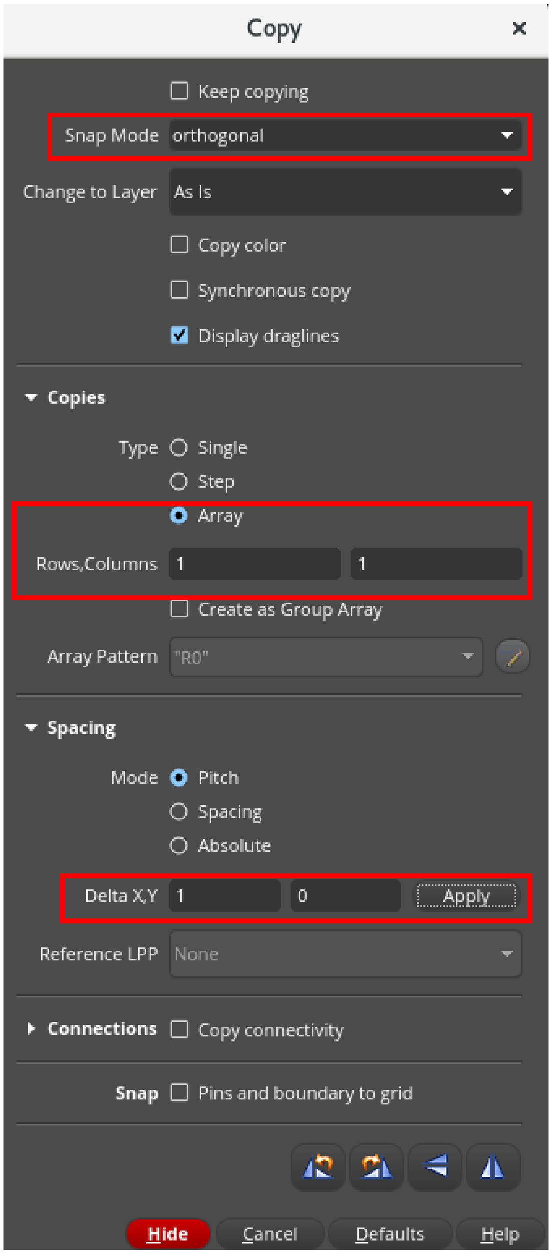
Using the *Copy* Command

1. Select the square in the layout canvas.

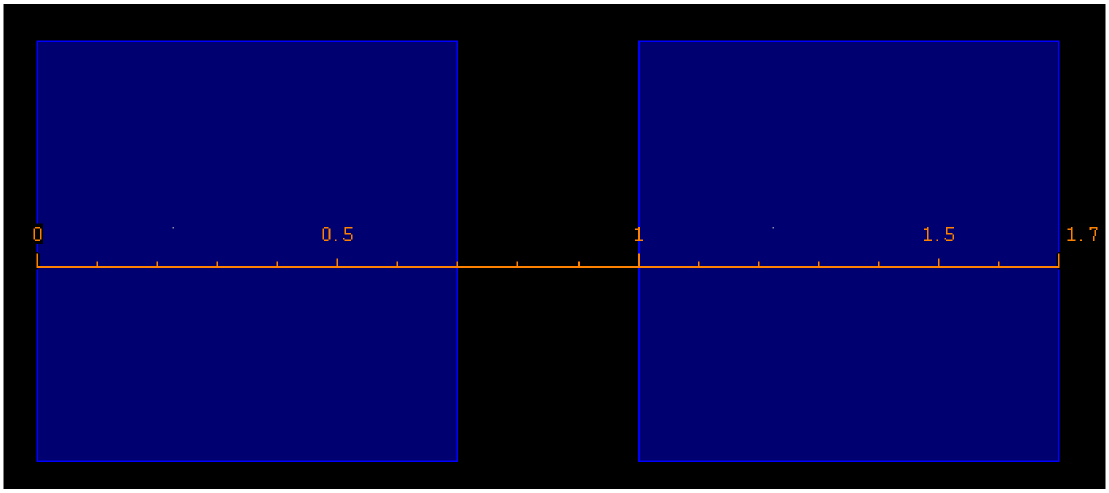
2. Choose **Edit – Copy** or click the **Copy** toolbar icon or press the bindkey **C** to open the Copy form.

   Note: If the Copy form is not visible, press **F3** to display the form.

3. In the Copy form, select the options, as shown here.
   - Snap Mode: orthogonal
   - Array: Rows: 1; Columns: 1
   - Delta: X: 1; Y: 0



4. Click **Apply** to make the copy, as shown here.



Using the *Repeat Copy* Command

Note: Multiple copies can be made using the *Repeat Copy* command.

1. In the layout window, delete all copies you may have made, leaving the original square.

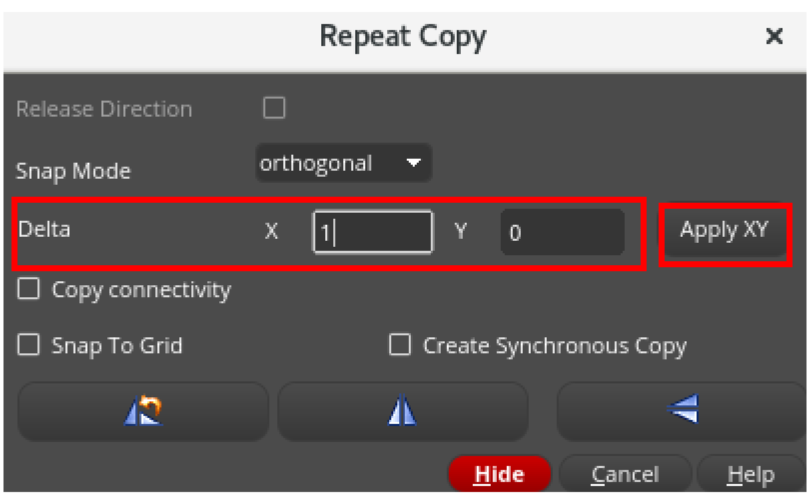
2. Reselect the square.

3. Choose **Edit** – **Repeat Copy** or click the **Repeat Copy** toolbar icon to open the Repeat Copy form.

   Note: If the Repeat Copy form is not visible, press **F3** to display the form.

4. In the Repeat Copy form, select the options, as shown here.

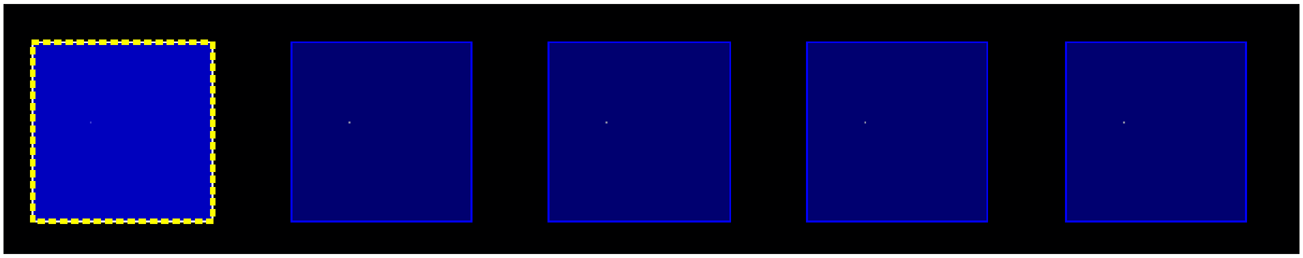
   - In the Delta field, change the X value to **1**.



5. Click **Apply XY** to make the copy.

   The copy is made once.

6. Press the **Period** (**dot**) key on your keyboard to continue making copies.

7. Click **Cancel** in the Repeat Copy form.

8. Close the design window. Do **NOT** save.

Using the *Copy connectivity* Option in the *Copy* Command

Note: When the *Copy connectivity* option in the *Copy* command is selected, the connectivity information on the path, pathseg, fluid guard ring, and via will be copied to the new object.

1. This is an optional exercise. Open the **test_cells/copyconnectivity/layout** view and explore the *Copy connectivity* option in the *Copy* command.

Using the *leRepeatCopyMoveStretch()* SKILL Function

Note: The **leRepeatCopyMoveStretch()** API enables you to repeatedly copy, move or stretch the selected objects. The command repeats the last ran command among the Copy, Move, and Stretch commands in preselect mode. The command uses the **dx** and **dy** from the last copy, move, or stretch operation.

1. This is an optional exercise. Open the **test_cells/repeatcopy_move_stretch/layout** view and explore the **leRepeatCopyMoveStretch()** SKILL® function.

2. Close any open forms and cellviews.

3. Leave the Virtuoso session.

4. In the shell, enter the following command:

   ```
   cd ~
   ```

   This command places you in your home directory.

Summary

In this lab, you learned how to

- Use the *Copy* and *Repeat Copy* commands

End of Lab

Module 3: Advanced Edit Commands

Lab 3-1 Sizing and Layer Generation

Objective: To examine the capabilities of sizing and layer generation.

In this lab, you use the *Layer Generation* command to change the layer and size.

Lab Setup

1. Follow the steps below to set up the environment and perform the lab:

2. If you continue from the previous lab, move to the sub-heading: Starting the session Otherwise, proceed to the next step.

3. Update the following line in the /VLPT2_IC_25_1/. cshrc_VLPT2 file with the appropriate valid local tool paths for Virtuoso®.

   ```
   setenv CDSHOME /ccstools/cdsind1/Software/IC251Base_lnx86
   ```

4. Save and close the file.

5. Source the updated class setup file.

   ```
   Source .cshrc_VLPT2
   ```

Starting the session

1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

   ```
   cd ./ M03_VLPT2_AdvEdit
   ```

2. If the Virtuoso session is not running, enter this command in the terminal window:

   ```
   virtuoso &
   ```

   The Command Interpreter Window (CIW) appears.

3. Create a new cellview: *ether/layGen/layout*

Library	ether
Cell	layGen
View	layout

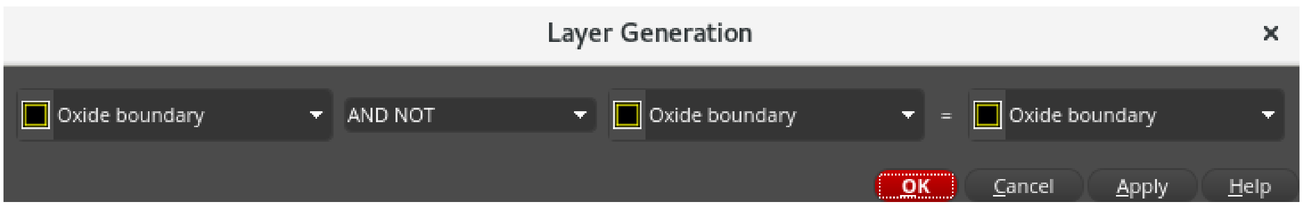
Note: The *ether/layGen_sample* cell is available for reference.

4. Create a **1x1** rectangle using **Oxide drawing**.

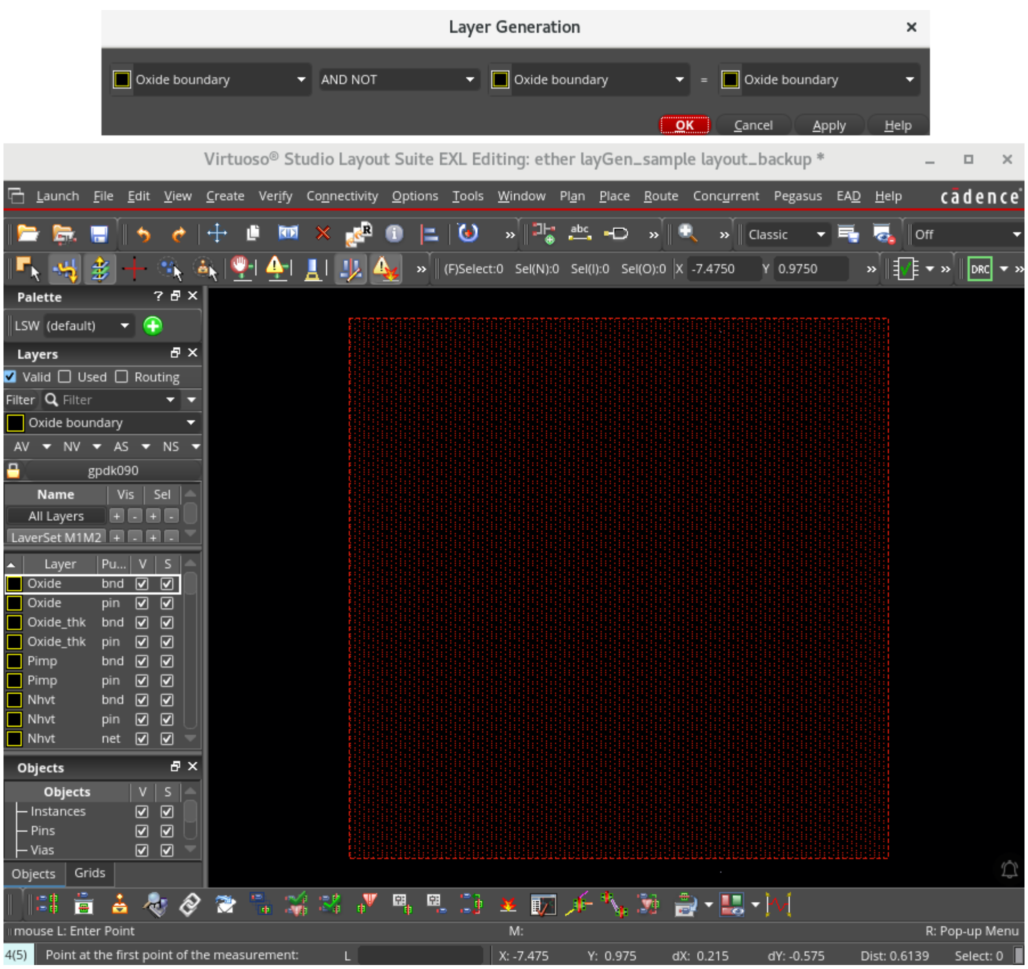
5. Select the rectangle.

   You will create an implant layer around the Oxide layer.

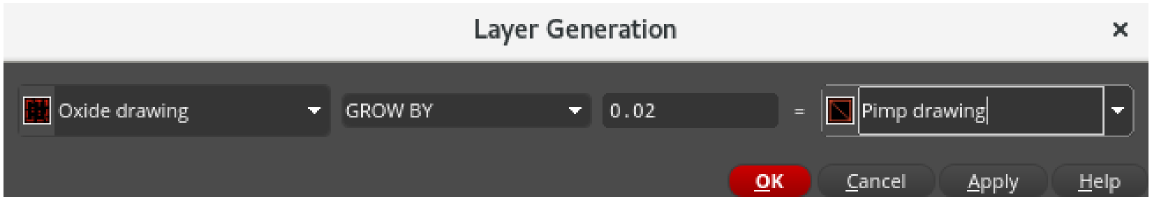
6. In the layout window, choose **Tools – Layer Generation** to open the Layer Generation form, as shown here.



7. Position the Layer Generation form above the *layGen* canvas so that it remains visible.

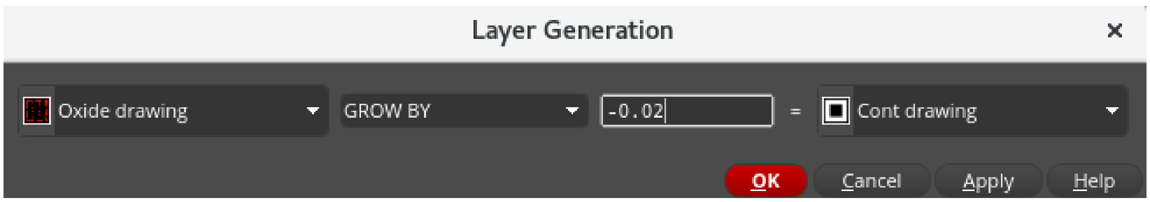


8. Modify the Layer Generation form, as shown here and click **Apply**.



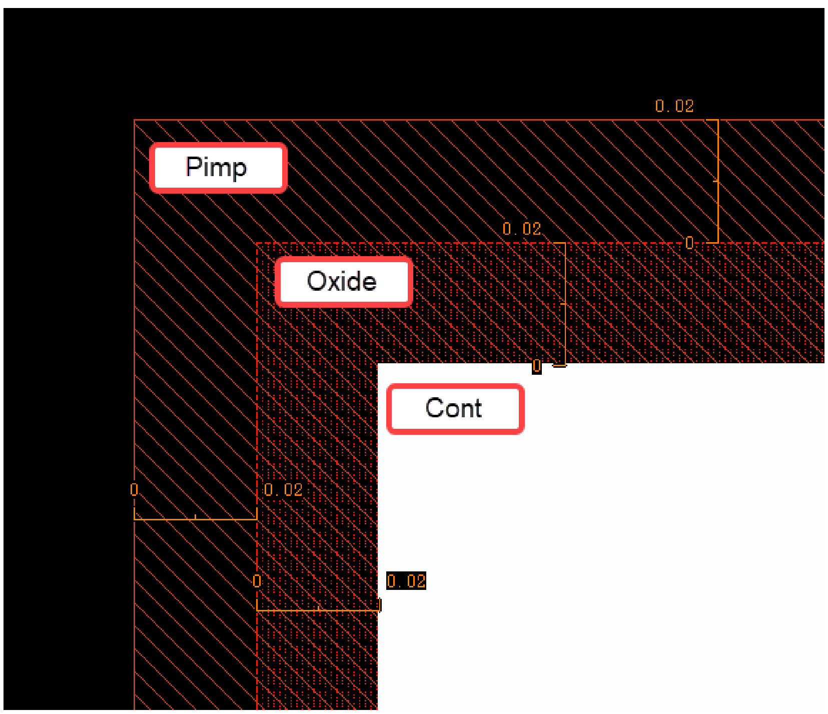
You see, the P-Implant is now visible enclosing the Oxide.

9. Change the form to create a *Cont* that is **-0.02** smaller than *Oxide*.



10. Click **OK**.

Zoom in to one of the corners, and you can observe that *Cont* is enclosed by the *Oxide* and *Pimp* layers, as shown here.



11. Close the cellview without saving.

12. Leave the Virtuoso session open for the next lab.

Summary

In this lab, you learned how to

- Use the *Layer Generation* command to change the layer and size

End of Lab

Lab 3-2 Using the Stretch Options

Objective: To use the various stretch options.

The **Stretch** command is a very powerful tool to help you edit and control objects in your design. It offers controls to keep wires connected between objects during a stretch so that you don't inadvertently create shorts or opens while editing.

1. In the CIW, choose **File – Open** and enter the following values in the form.

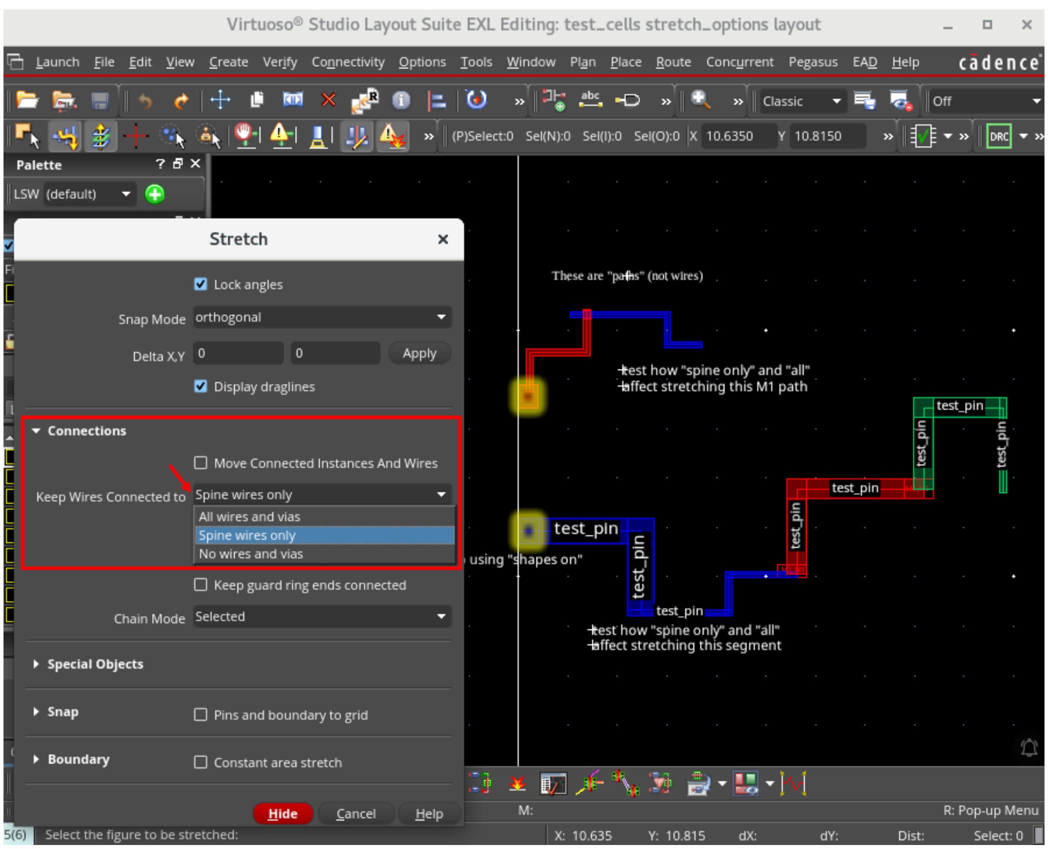
Library	`test_cells`
Cell	`stretch_options`
View	`layout`

2. Click **OK**.

   The *test_cells/stretch_options/layout* cellview opens for editing.

3. In the layout, choose **Edit – Stretch** or click the Stretch toolbar icon or press the bindkey **S** to open the Stretch form.

   Note: Press **F3** to bring up the Stretch form if it is not already up.



4. Use the wires in the layout view to test the various Stretch options:

   a. Stretch with **Keep Wires Connected to**: *Spine wires only* (default)

   b. Stretch with **Keep Wires Connected to**: *All wires and vias*

   c. Stretch with **Keep Wires Connected to**: *No wires and vias*

   Notice that in the layout cell, there are also "normal" paths. Test using the different **Stretch** options on these as well.

5. Choose **File – Close** to close the layout. Discard any edits.

6. Summary Close any open forms and cellviews.

   a. Leave the Virtuoso session.

   b. In the shell, enter the following command:

   c. cd ~

   This command places you in your home directory.

Summary

In this lab, you learned how

- The different stretch options affect working with paths and path segments

End of Lab

Module 4: Hierarchical Editing

Lab 4-1 Using the Search, Save/Restore, and Navigator Assistant

Objective: To use the Search, Save/Restore, and the Navigator options.

In this lab, you use the Search assistant to locate instances, store them in a selection set, create a group, and use an existing layout as a guide to update a layout design.

The original design is *ether/trans_NOR_1X/layout*, and the layout you will update is *ether/trans_NOR_1X/layout_vls*. The new layout is similar to the original layout, except that the height of the new layout is reduced from **8µm** to **7.5µm** to reduce the wire length between NMOS and PMOS. Your task is to place the PMOS devices in the new layout, similar to the placement in the original layout.

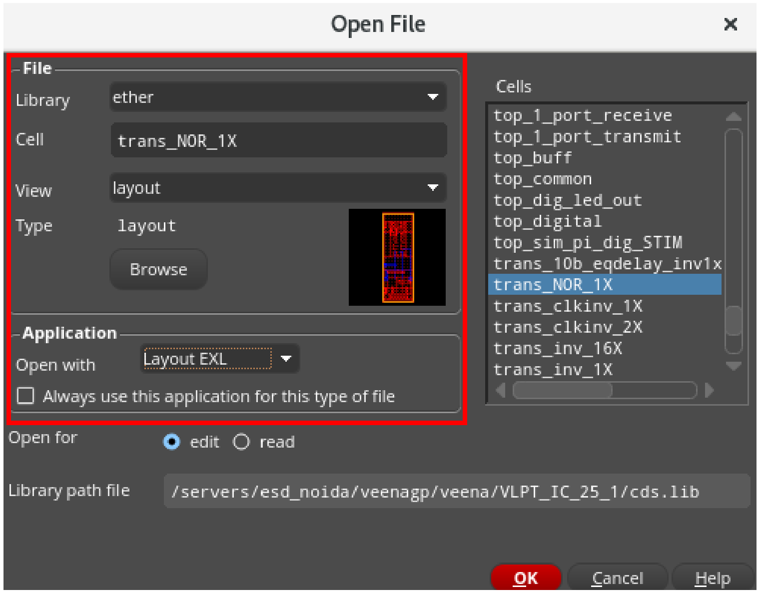
Lab Setup

1. Follow the steps below to set up the environment and perform the lab:
2. If you continue from the previous lab, move to the sub-heading: Starting the session Otherwise, proceed to the next step.
3. Update the following line in the /VLPT2_IC_25_1/. cshrc_VLPT2 file with the appropriate valid local tool paths for Virtuoso®.
   ```
   setenv CDSHOME /ccstools/cdsind1/Software/IC251Base_lnx86
   ```
4. Save and close the file.
5. Source the updated class setup file.
   ```
   Source .cshrc_VLPT2
   ```

Starting the session

1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:
   ```
   cd ./ M04_VLPT2_HierarchicalEdit
   ```
2. Invoke the Virtuoso software by entering this command in the terminal window:
   ```
   virtuoso &
   ```

3. Open the following layout view.



The *ether/trans_NOR_1X/layout_vls* cellview opens for editing.

Note: The view is *layout_vls*.

In the next steps, you will see some configurations for opening, positioning, and using the Search assistant.

Using the Search Assistant and the Save/Restore Option

1. In the layout canvas, choose **Window – Assistants – Search**.

   The Search assistant appears on the right side of the canvas.

2. Float the **Search** assistant outside the right side of the canvas.

3. Click the **X** button on the top right of the Search assistant to close the assistant.

4. Again, choose **Window – Assistants – Search**.

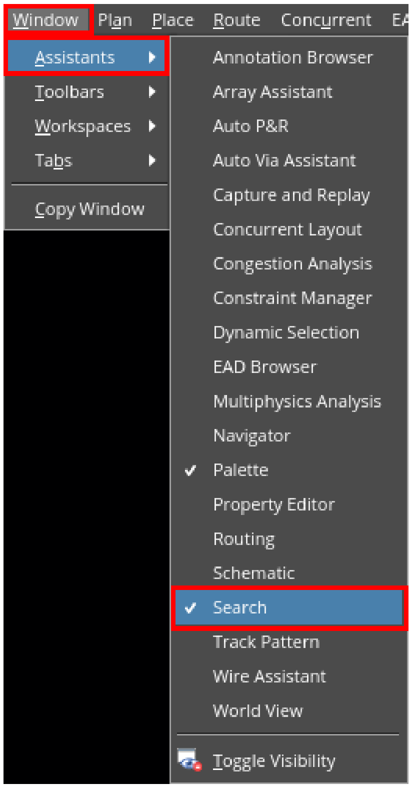
   The Search assistant is presented outside the canvas where you had previously positioned it.

5. Move the **Search** assistant so that it partially covers the right side of the canvas and release the **left** mouse button.

   The placement location for the assistant will open based on the cursor crossing the canvas boundary.

6. Close the **Search** assistant.

7. **Right**-click in the toolbar in the empty area just above the toolbar and enable the **Search** assistant, as shown here.



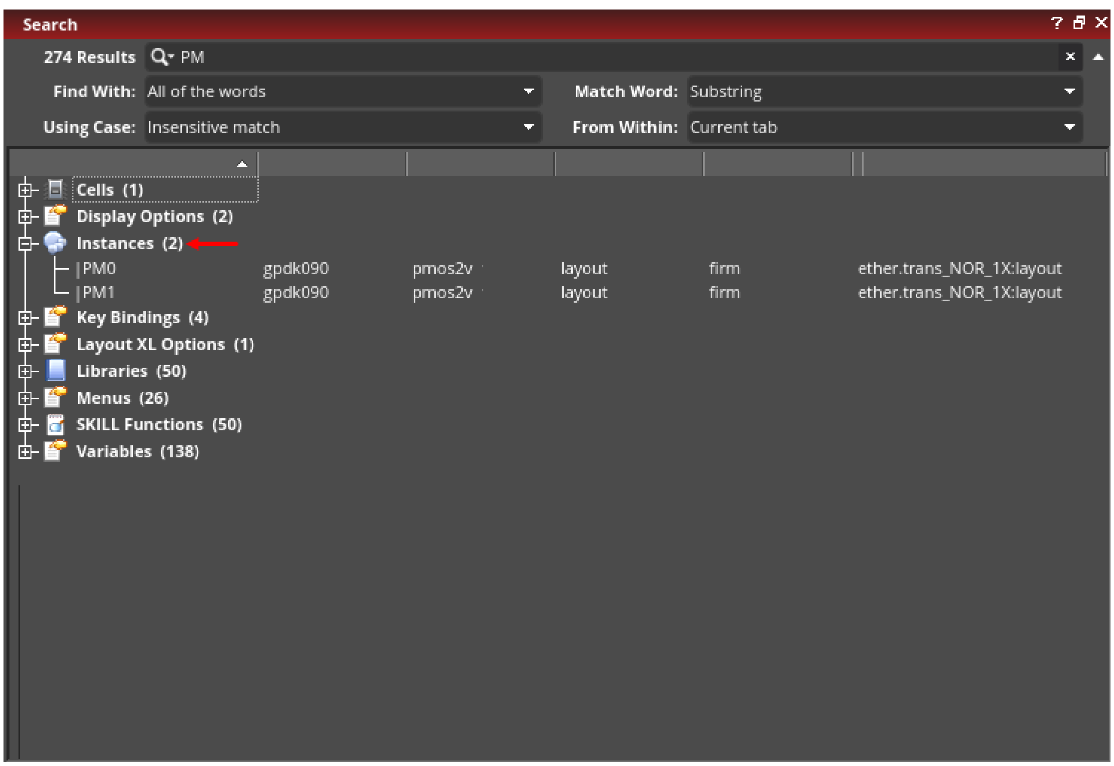
Note: You previously started the Search assistant by selecting **Window – Assistants – Search**.

8. In the Search assistant, enter **PM** in the search field and press **Enter**.

The search results appear, as shown here.

9. Expand the **Instances** heading in the returned search results if it is not already expanded.

   You see two instances in the design: **|PM0** and **|PM1**.

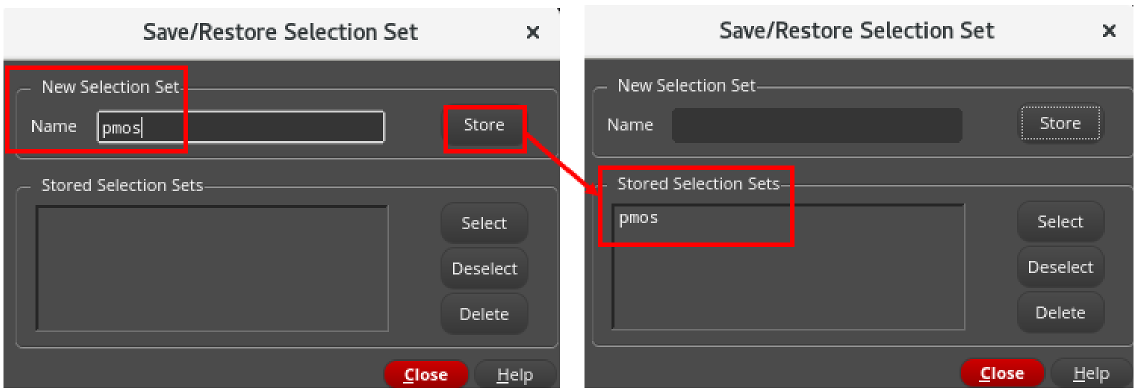


10. Select these two instances **|PM0** and **|PM1** in the Search assistant. They will be selected in the layout window as well.

11. Save this selection set for later use.

    a. In the layout window, choose **Edit – Select – Save/Restore**.

    b. In the Save/Restore Selection Set form, specify the *Name* of the selection set as **pmos,** as shown here.



    c. Click **Store** and **close** the form.

    Note: If you get a WARNING in the CIW that *pmos Selection Set* already exists, delete it and click **Store** again.

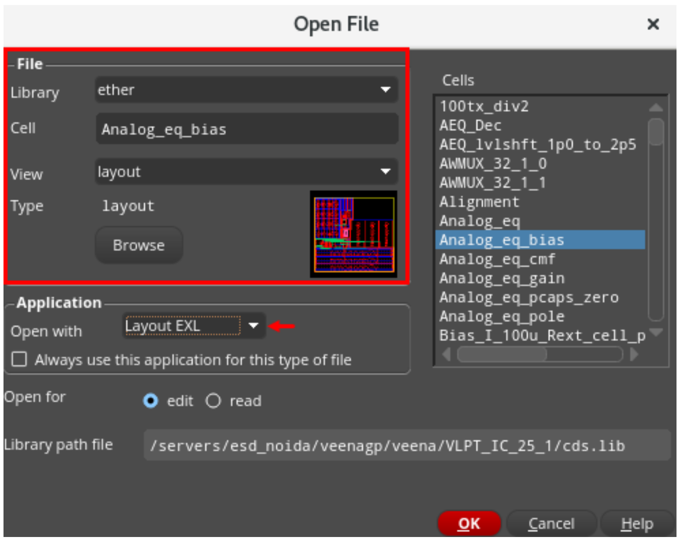
12. Press the **Ctrl+D** keys in the layout design to deselect everything.

13. Leave the cellview open.

Using the Navigator Assistant

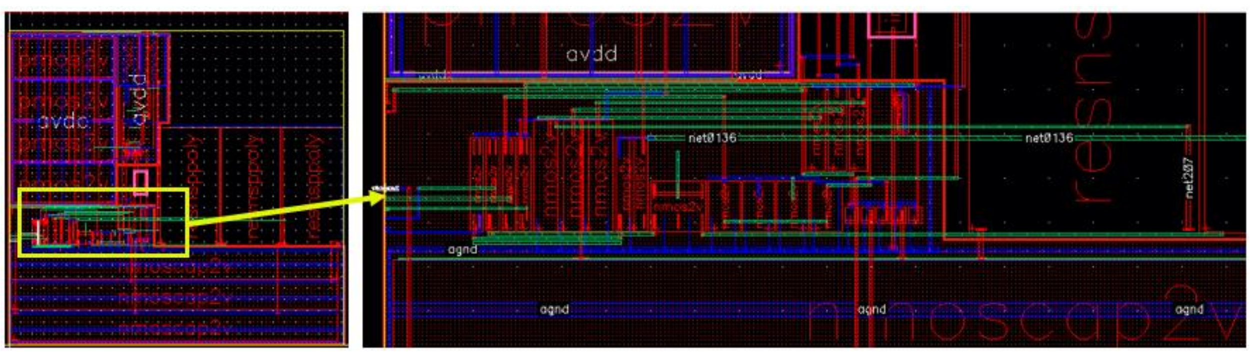
The Navigator lists the design objects in the cell. You can use it to locate and identify objects. Items selected in the Navigator will have their properties displayed in the Property Editor assistant.

1. From the opened layout, choose **File – Open** and open the following layout view.



Note: The *trans_NOR_1X* and *Analog_eq_bias* cellviews are available as tabs.

2. Zoom in to the left middle of the *Analog_eq_bias layout* cellview, as shown here.



3. Choose the *workspace* as **Basic** in the layout design.

4. Select one of the MOS devices in this area.

   Note: The Navigator highlights the selected device and provides the Property Editor information for the device.

5. In the Navigator assistant, set the OBJECTS category to **Nets**.

   a. Move down the list and select the ***I_rext_100_3*** net.

See how it highlights the net in the layout canvas.

6. In the layout canvas, choose **View – Zoom To Selected** or press the bindkey **Ctrl+T**.

   The canvas zooms in to the location of the pin.

7. In the layout window, choose **View – Zoom Out** or press the bindkey **Shift+Z** a couple of times to zoom out.

8. Click on the **trans_NOR_1X** tab **or** open the ***ether trans_NOR_1X layout_vls*** *cellview*.

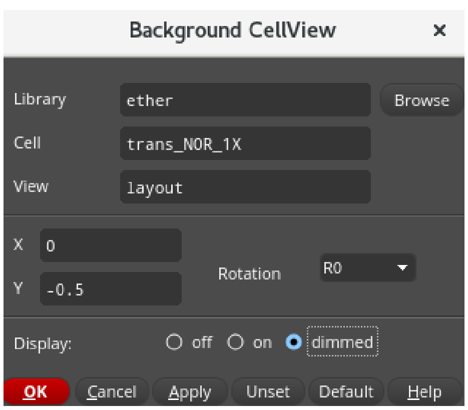
Overlaying a Cell to Use as a Template

1. Choose **View – Background** in the *ether/trans_NOR_1X/layout_vls* design window.

2. In the Background CellView form, enter the name of the *ether/trans_NOR_1X/layout* cellview as a background at X = 0 and Y = -0.5.

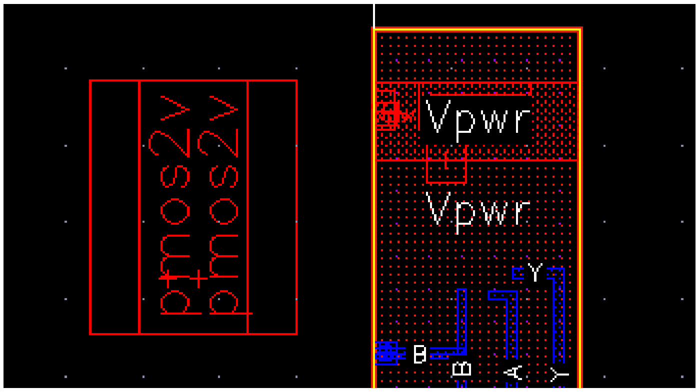
   The reason for **Y = -0.5** is that the height of the design was reduced from **8µm** to **7.5µm**.

   The form looks as shown here.



3. Click **OK** in the Background CellView form.

   The background cellview appears as an instance. The PMOS instances in the background cellview are dimmer than the cellview being edited.

   You will move the **|PM0** and **|PM1** instances so that they exactly overlap the corresponding instances in the background cellview.

   Because there are many layers visible in the layout, it may be difficult to visually align the instances. If you can see just the metal and contacts of the PMOS instances, that should help in reducing the visual clutter.

   Tip: To quickly change the visible layers, use the Layers Assistant AV NV AS NS.

4. In the layout window, press **Shift+F** to view all the levels of the hierarchy.

5. In the Layers assistant, enable **Used Layers Only**.

6. Click on the ***Metal1 drawing*** layer.

7. Use the **middle** mouse button and drag from the top to bottom of the list of Used Layers.

   This will turn off the visibility and selectability for all the layers. The entry layer will remain visible. You could have also clicked the **NV** (None Visible) button.

8. Use the **middle** mouse button to make the **Cont drawing** visible.

9. The display updates to show the **Metal1 drawing** and **Cont drawing** layers in the design window.

   The next step is to move the PMOS instances.

   The PMOS instances **|PM0** and **|PM1** are abutted. You have to be careful not to break the abutment. You should also be careful not to move other objects accidentally. To do this, you will create a figGroup out of the **|PM0** and **|PM1** instances and make all the layers and objects except group unselectable.

10. Select the two PMOS devices to the left of the cellview.

11. In the layout canvas, choose the **Create – Group** menu command or use the **Create Group** icon in the Edit toolbar to create a group of the selected objects.

    You have created a group named *Group0* for the PMOS instances **|PM0** and **|PM1**.

12. In the Objects assistant, turn off the selectability for *Shapes*, *Boundaries*, *Soft Blocks*, and *Blockages*.

13. Turn on the **Visible** and **Selectable** selection buttons for *Fig Groups*.

    You can now select only the group objects.

14. Using the mouse, move the group to exactly overlap the contacts in the background design.

    You can change the snap mode to **anyAngle** to do this.

15. After you are satisfied with the result, ungroup the PMOS group by selecting the **Edit – Group – Ungroup** menu command or use the **Ungroup** icon in the Edit toolbar and point at the group to ungroup it.

16. In the Layers assistant, turn on the visibility and selectability of all the layers.

17. In the Objects assistant, turn on the selectability for *Shapes*, *Boundaries*, *Soft Blocks*, and *Blockages*.

18. Turn off the display of the background cellview by choosing **View – Background**.

19. In the Background CellView form, set the **Display** radio button to **off**.

20. Click **OK** in the Background CellView form.

21. Save the modified design, *ether/trans_NOR_1X/layout_vls*, using **File – Save** in the viewport.

    Note: The *ether/trans_NOR_1X/layout_vls_sample* cellview is available for reference.

22. To confirm that you have made correct updates, select either **|PM0** or **|PM1** and bring up the Edit Properties form.

    The Y coordinate should show 4.24, which is 0.5 less than the instances in *ether trans_NOR_1X layout*.

23. Close the layouts.

24. Leave the Virtuoso session running for the next lab.

Summary

In this lab, you learned how to

- Use the Search, Save/Restore, and the Navigator options

End of Lab

Lab 4-2 Working with Groups

Objective: To use the Group and Ungroup commands.

In this lab, you use the ***Group*** and ***Ungroup*** commands to contain the design objects so that they may be moved or aligned in the same configuration.

The *Group* command lets you group shapes and/or instances together for you to manipulate the objects as a whole rather than individually. It will have as an owner an area boundary object, which will conform to the bounding box of the entire group (the composite of all the bounding boxes of the members of the group). This area boundary is automatically created, edited, and deleted by the ***Group*** command as objects are added to or deleted from the group.

A group:

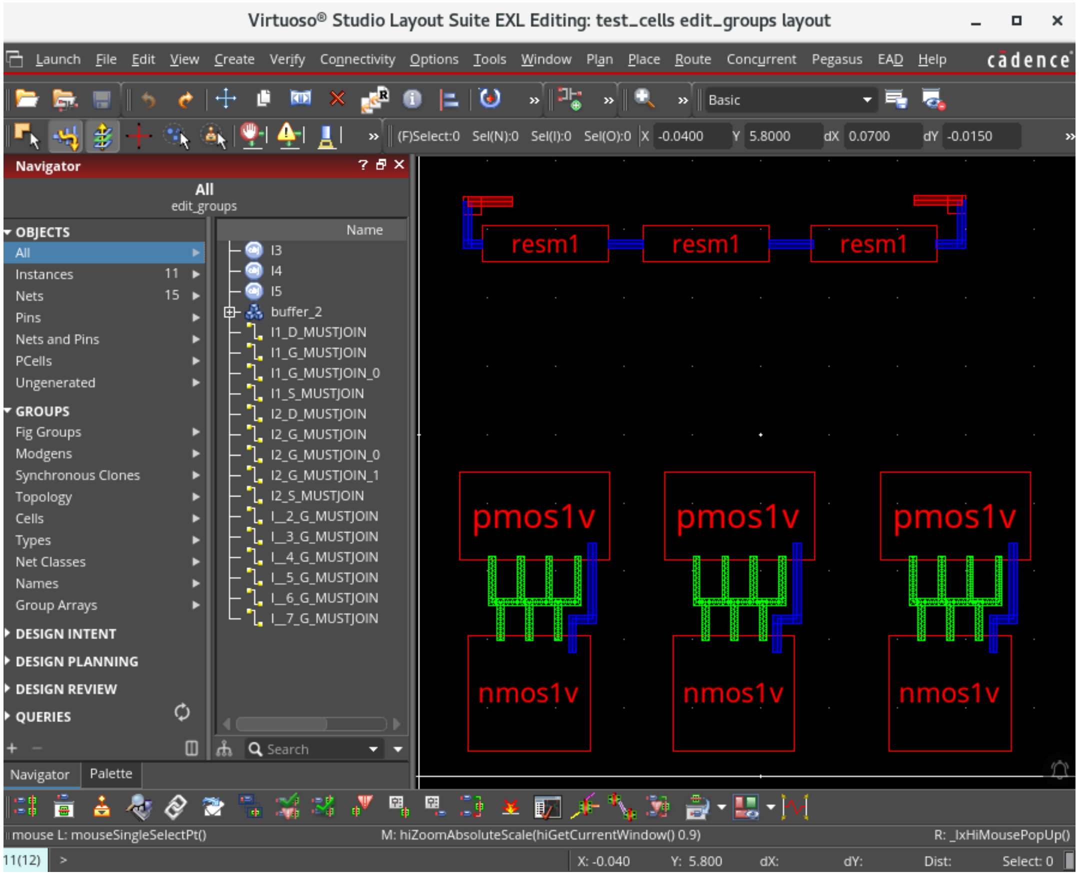
- Is a named database figure that you can query in the database.
- Can contain any number of instances, including other groups.
- Is selectable.
- Has an origin and an orientation.
- Does not have a layer/purpose, though its members may.
- Does not have connectivity, though its members may.
- May have an applied constraint group.

1. In the CIW, choose **File – Open** and enter the following values in the form.

Library	`test_cells`
Cell	`edit_groups`
View	`layout`

2. Click **OK**.

   The *test_cells/edit_groups/layout* cellview opens for editing.



Exploring *Group* and *Ungroup* Commands with Instances

1. Hover your cursor over any of the Instances.

   You will notice that all four Instances are surrounded by a dotted bounding box.

   This is an indication that all four Instances are members of a group.

2. Select any member of the group.

   All four Instances are now selected and can be moved or stretched, or deleted as a single component.

3. In the layout canvas, choose the **Edit – Group – Ungroup** menu command or use the **Ungroup** icon in the Edit toolbar.

   This will begin to ungroup the collective group structure, one group at a time.

4. Hover your cursor over the two left Instances and then to the two right Instances.

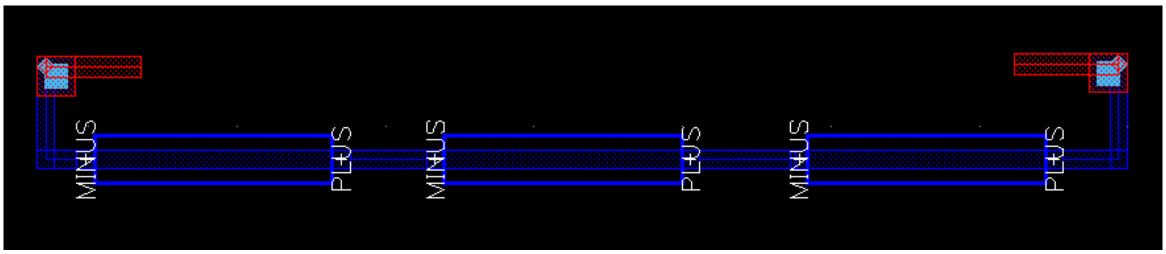
   You will notice that they have now been reduced to two individual groups.

5. You can continue ungrouping until you are left with four individual Instances.

   If you wish to create a group from Instances and interconnect, you must area-select all objects you wish to include in the group.

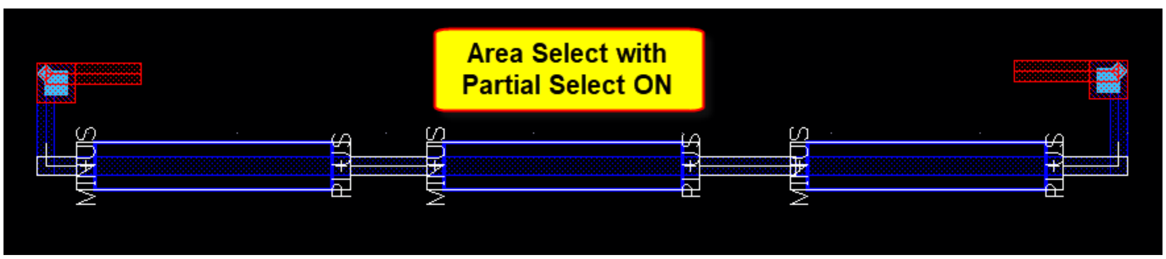
Exploring *Group* and *Ungroup* Commands with Resistors

1. **Pan** up to the three resistors in the layout canvas.



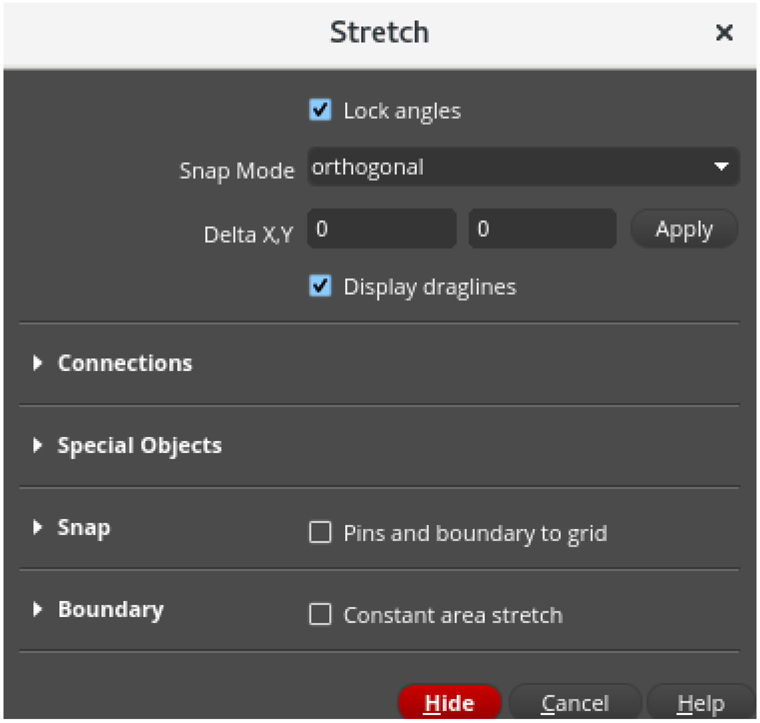
   For the resistors and their interconnects to behave like a group, you must area-select them first.

2. Area-select the three resistors, as shown here.



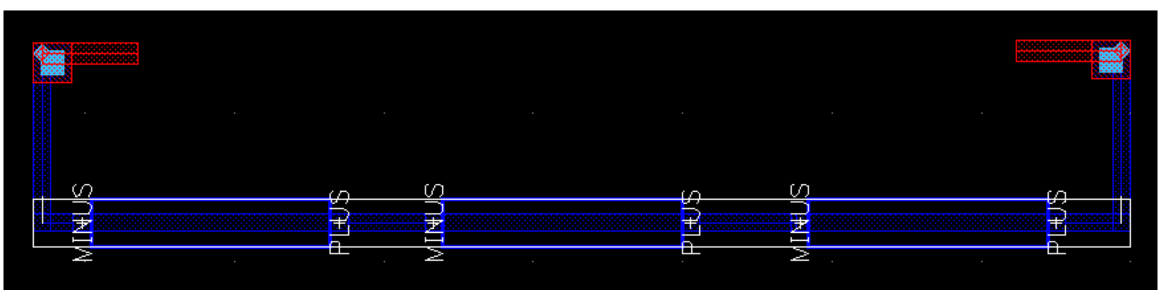
3. In the layout window, choose the **Create – Group** menu command or use the **Create Group** icon in the Edit toolbar to create a group of the selected objects.

4. In the layout design, choose the **Edit – Stretch** menu entry or use the **Stretch** icon in the Edit toolbar or press the bindkey **S** to invoke the Stretch command, as shown here.



5. Stretch the group downward.

   Notice that all three resistors behave as one, and interconnect at each end remains connected.



6. Explore ungrouping the created group.

7. Close the cellview. Do **NOT** save.

8. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you learned how to

- Use the *Group* and *Ungroup* commands

End of Lab

Lab 4-3 Searching the Database

Objective: To use the Find/Replace command and the Search assistant to find the labels in the design.

A quick way to explore the structures contained in your design is to use the *Find/Replace* command and the Search assistant. With these options, you can selectively search and identify the components in your design.

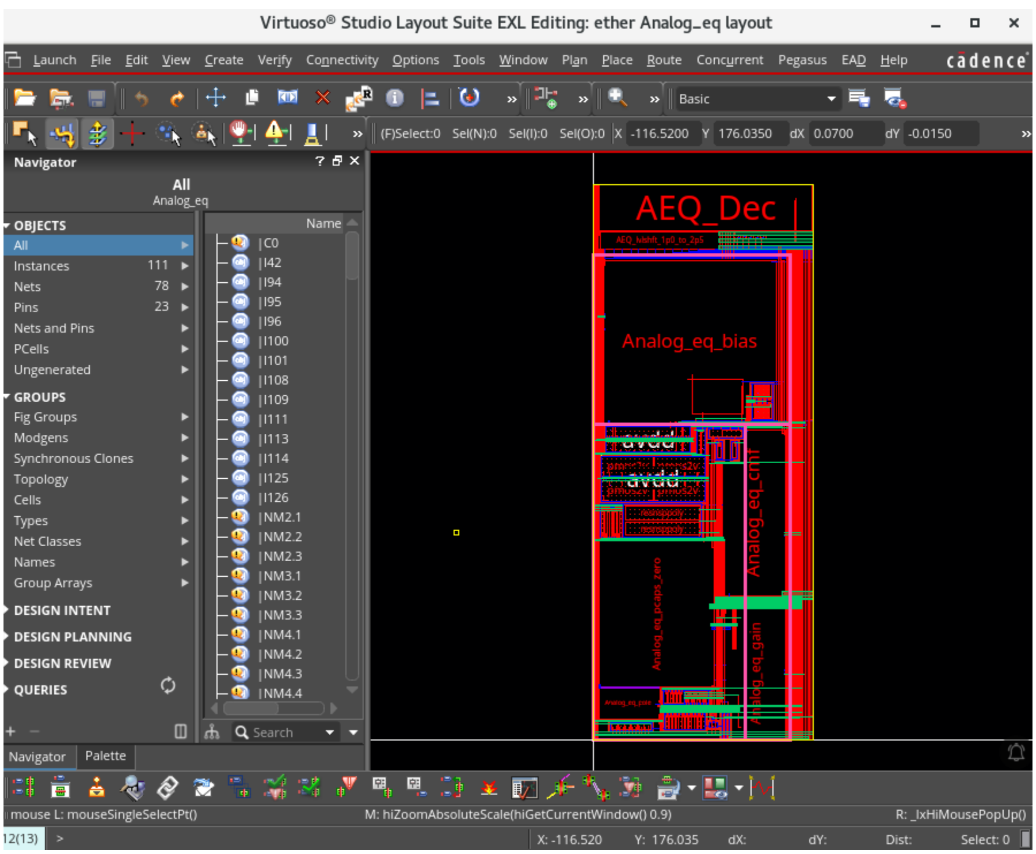
1. In the CIW, choose **File – Open** and enter these values in the form.

Library	`ether`
Cell	`Analog_eq`
View	`layout`

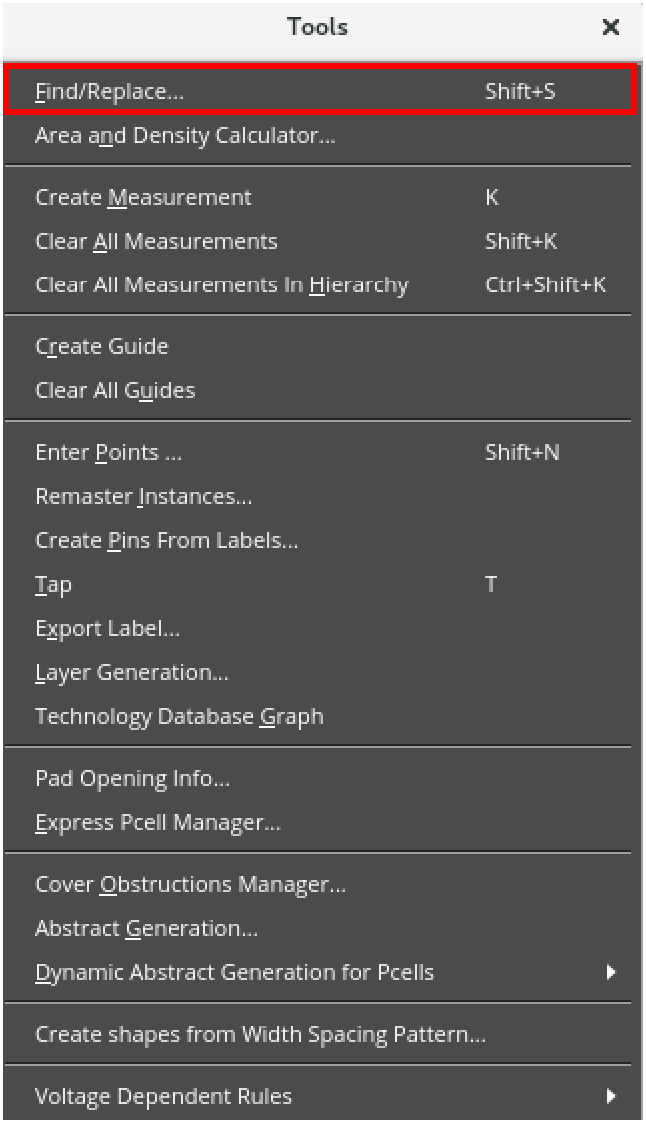
2. Click **OK**.

   The *ether/Analog_eq/layout* cellview opens for editing.

3. Set the *workspace* to **Basic** in the layout canvas, as shown here.

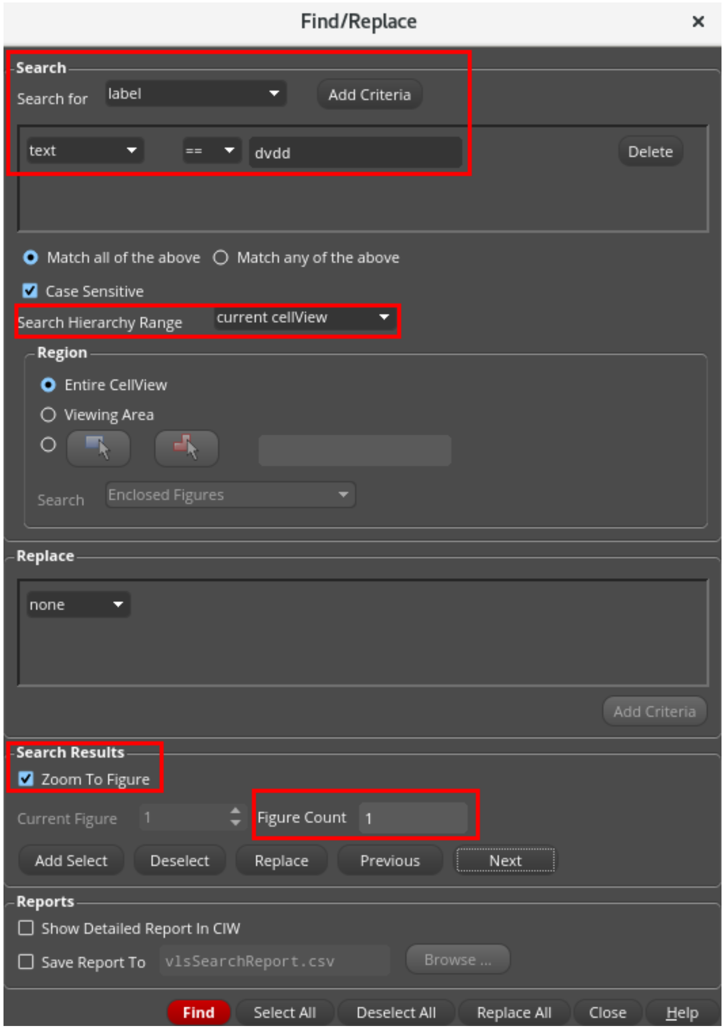


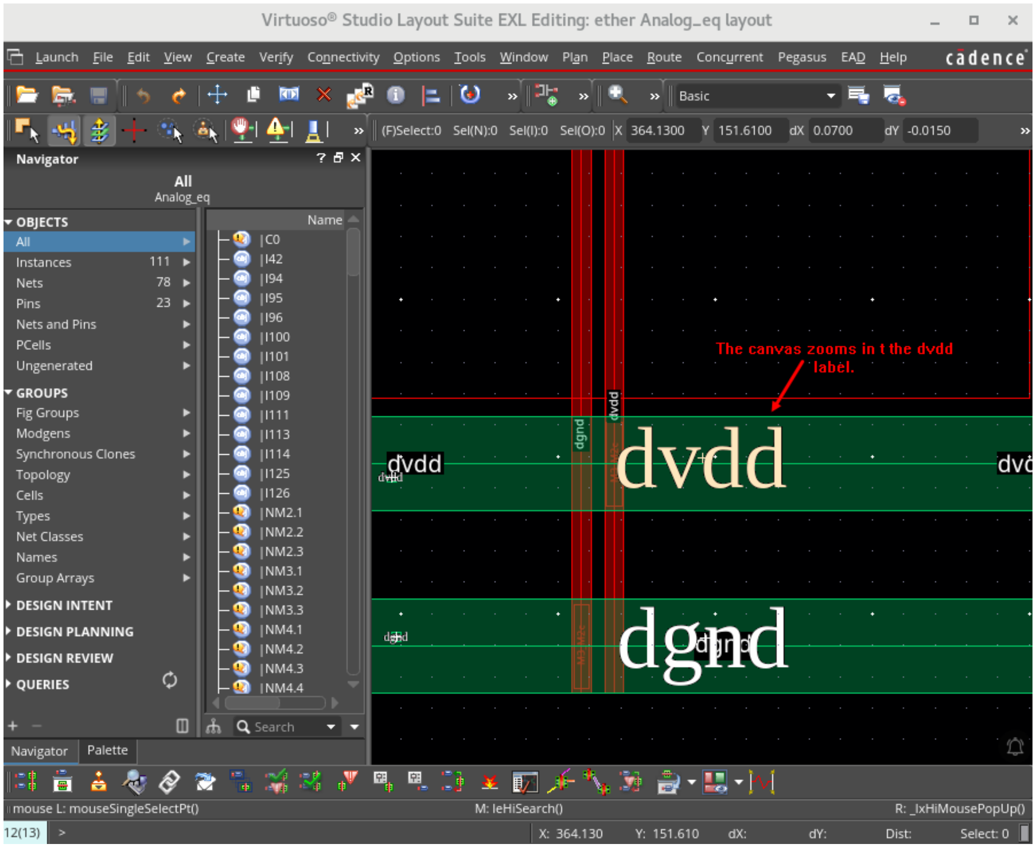
4. From the layout design window, choose the **Tools – Find/Replace** menu entry or press the bindkey **Shift+S** to open the Find/Replace form, as shown here.



5. In the Find/Replace form, select the options, as shown here.
   - Search for: **label**
   - Add Criteria: **text == dvdd**
   - Search Hierarchy Range: **current cellView**
   - Zoom To Figure: **Enabled**

6. In the Find/Replace form, click the **Find** button.
   a. The canvas zooms in to the **dvdd** label, as shown here.





How many labels are found?

Answer: The *Figure Count* field tells you how many labels were found during the search; in this case, it is **1**.

7. Close the *Find/Replace* form.

8. Press **F** to fit the cellview in the canvas.

9. In the layout window, choose **Window – Assistants – Search**.

   a. The Search assistant is now available on the right side of the canvas.

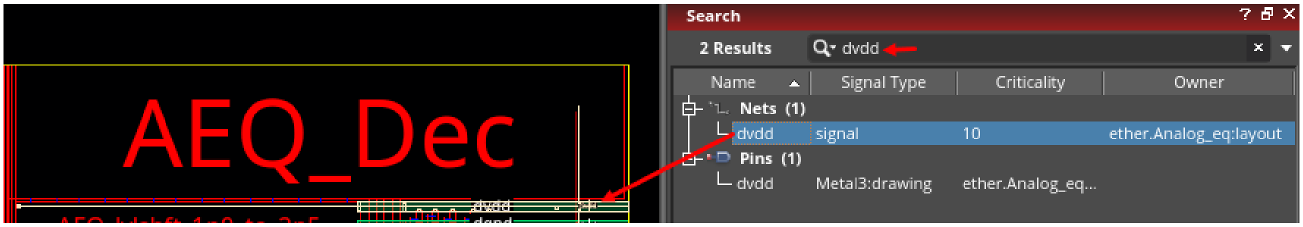
10. Enter **dvdd** in the *Search* field, and press **Enter** to execute the search.

    a. The *dvdd* nets and pins are found.

11. Click on the **Nets – dvdd** signal.

    a. The net is highlighted in the layout canvas and in the Navigator assistant.

12. In the layout window, choose the **View – Zoom To Selected** menu command or press the bindkey **Ctrl+T** to locate the highlighted net.



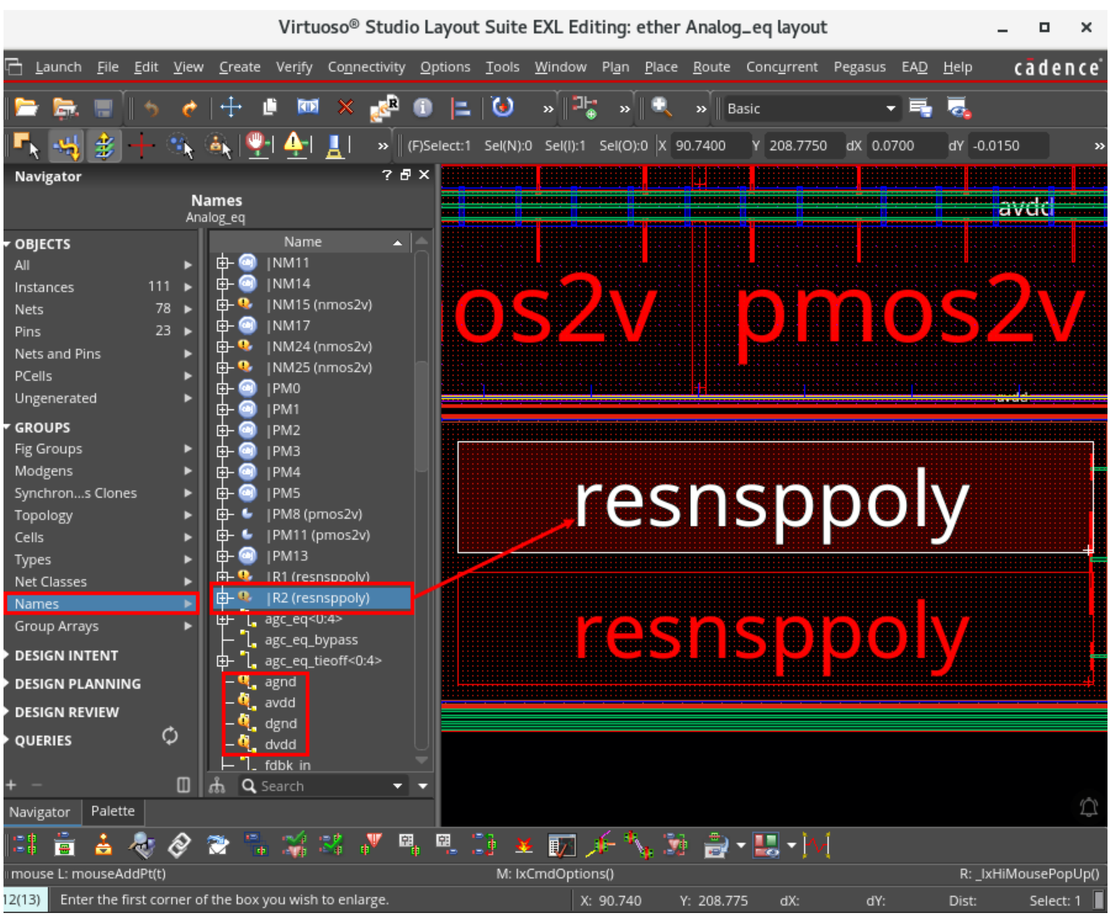
Note: There are **10** different highlight colors. If the highlight is not easy to see, click on the **Nets (1)** and then back to the **dvdd** signal. It will change the highlight color to the next available color.

13. Close the Search assistant.

14. In the layout window, choose the **View – Zoom To Fit All** menu command or press the bindkey **F** to fit the cellview in the canvas.

15. In the Navigator assistant, from the **GROUPS** category, select **Names** to highlight other nets and instances.

    a. Here, we have clicked on **|R2** (**resnsppoly**) and zoomed to that instance.



16. Select each of these to exercise the ***Navigator Names*** option.

    `agnd, avdd, dgnd, dvdd`

17. Choose **View – Zoom To Selected** (bindkey **Ctrl+T**) to locate the highlighted net.

18. **Close** the design window without saving.

Using the *Find/Replace* Enhancements

The *Find/Replace* utility has been enhanced to operate on a specific region. You can choose between *Entire CellView*, the current *Viewing Area*, or *a user-drawn rectangle/polygon region* for search. Smaller search region would reduce the search time and the number of search results.

1. This is an optional exercise. Open the **test_cells/find_replace/layout** view and explore the *Find/Replace* enhancements.

2. Leave the Virtuoso session running for the next lab.

Summary

In this lab, you learned how to

- Use the *Find/Replace* command and the Search assistant to find the labels in the design

End of Lab

Lab 4-4 Remastering Instances

Objective: To use the Remaster Instances command to update the layout.

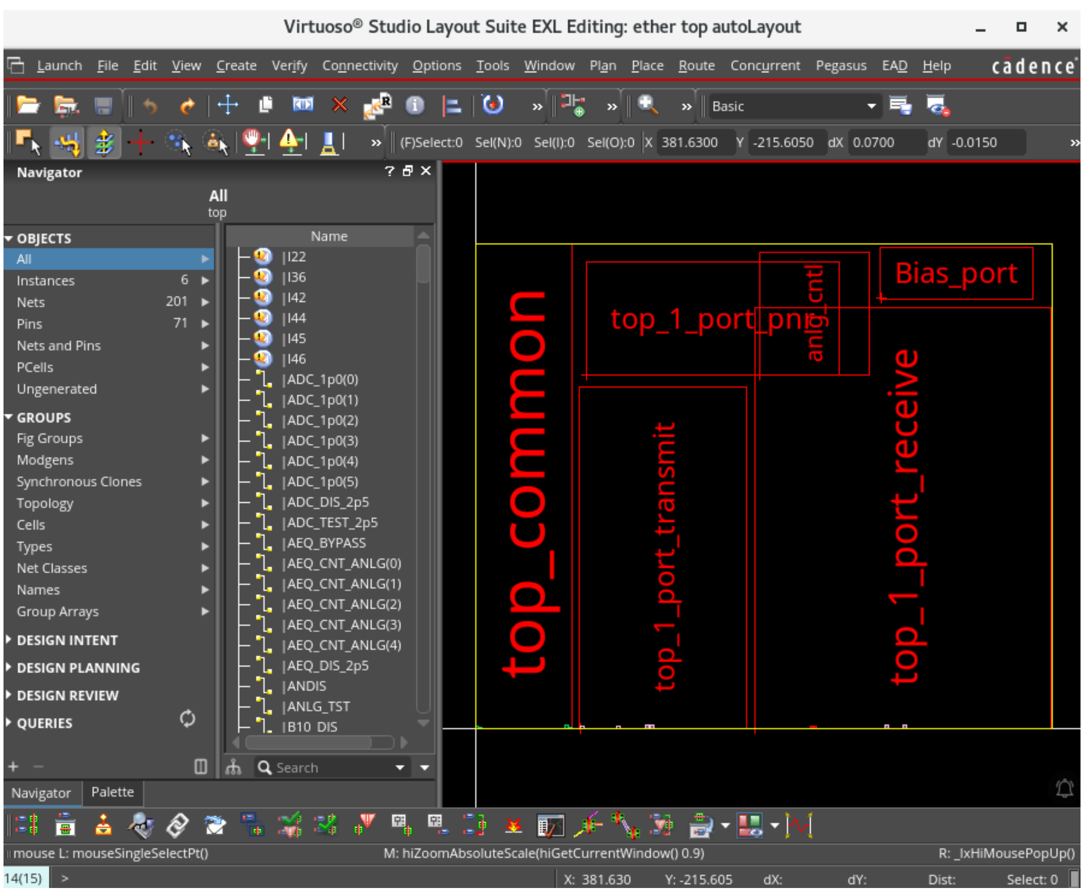
In this lab, you use the *Remaster Instances* utility to change the masters of the instances in your design. When a DEF file is imported into the Virtuoso Design Environment, the top layout will have instances of abstracts in the design. The *Remaster Instances* command can be used to replace the abstracts with a layout view.

1. In the CIW, choose **File – Open** and enter the following values in the form.

Library	ether
Cell	top
View	autoLayout

2. Click **OK**.

   The *ether/top/autoLayout* cellview opens for editing.

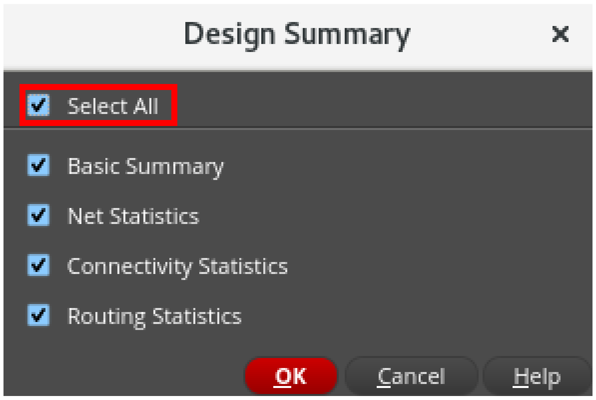


Exploring the *Remaster Instances* Utility

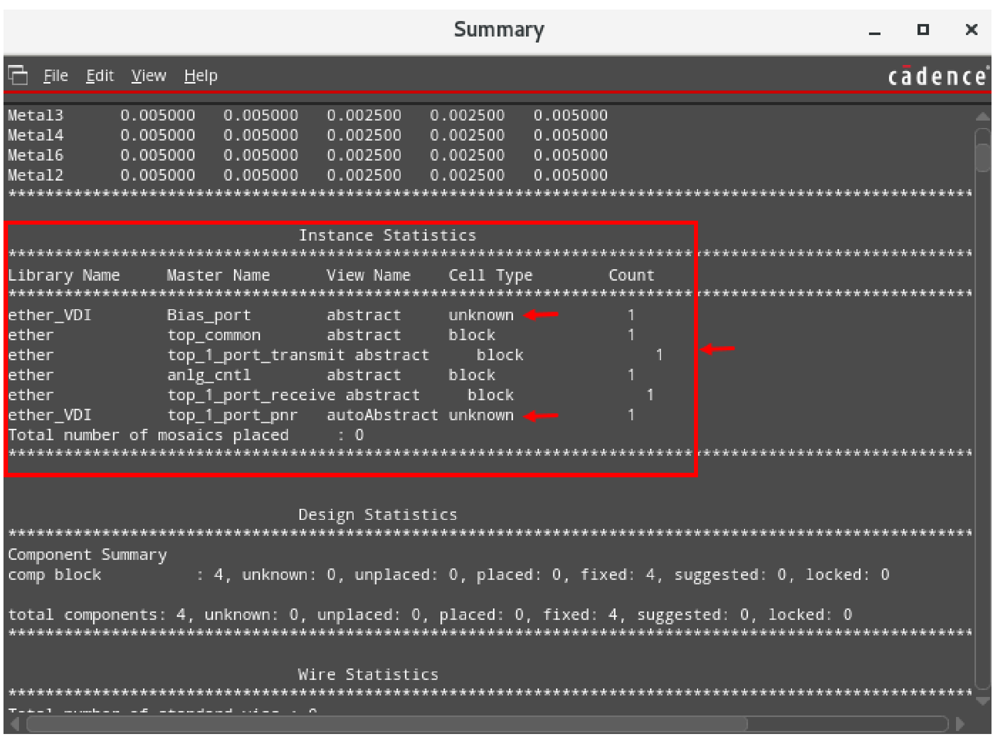
1. In the layout canvas, choose **File – Summary** to open the Design Summary form.

   a. In the Design Summary form, click the **Select All** button.

   b. Click **OK** to open the *Summary* window.



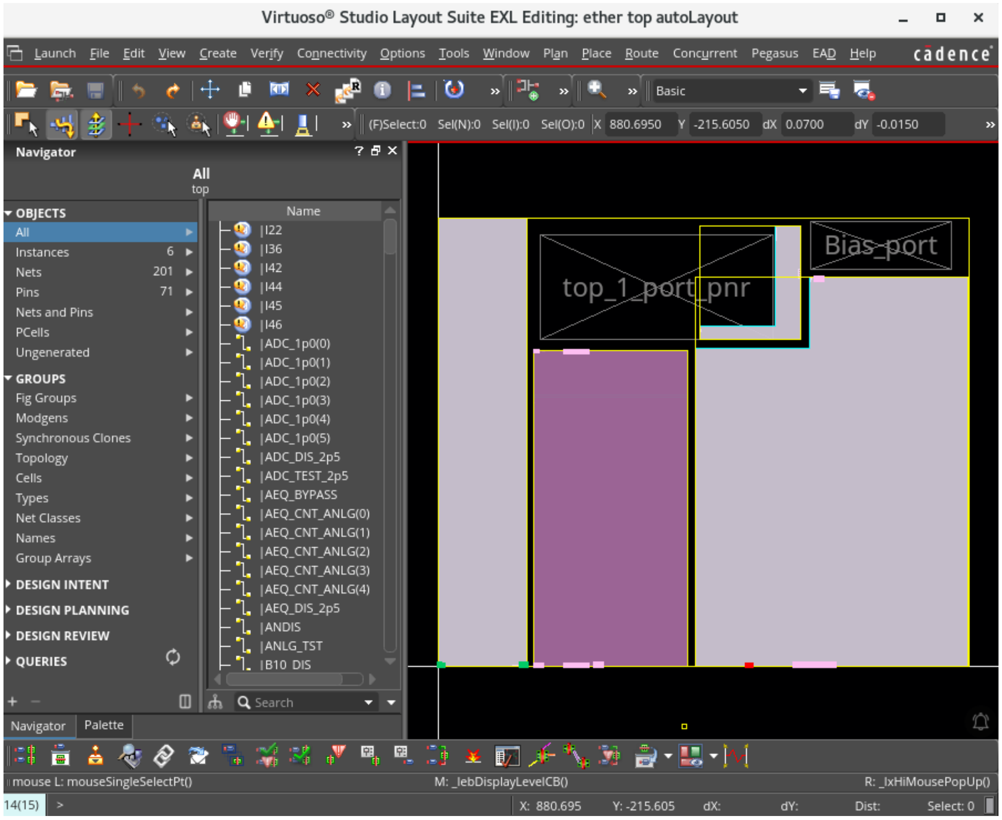
In the *Summary* window, a summary listing appears with *Instance Statistics* showing the cells referenced in the design. Look through the *Instance Statistics* to see the view names are abstract. The *ether_VDI/top_1_port_pnr/autoAbstract* and *ether_VDI/Bias_port/abstract* have unknown *Cell Type* as shown here.



2. Choose **File – Close Window** to close the *Summary* window.

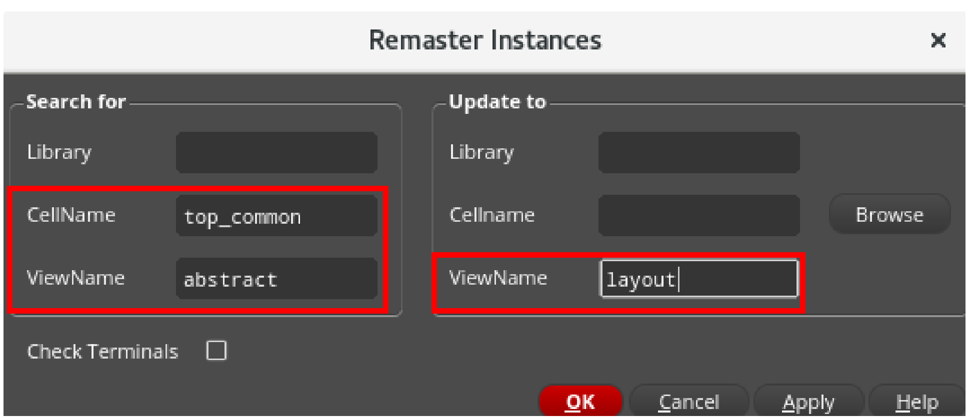
3. In the design window, press **Shift+F** to view all levels of the hierarchy.

The ether_VDI/top_1_port_pnr/autoAbstract and ether_VDI/Bias_port/abstract have flashing markers, as shown here.



4. In the layout canvas, choose the **Tools – Remaster Instances** menu command.

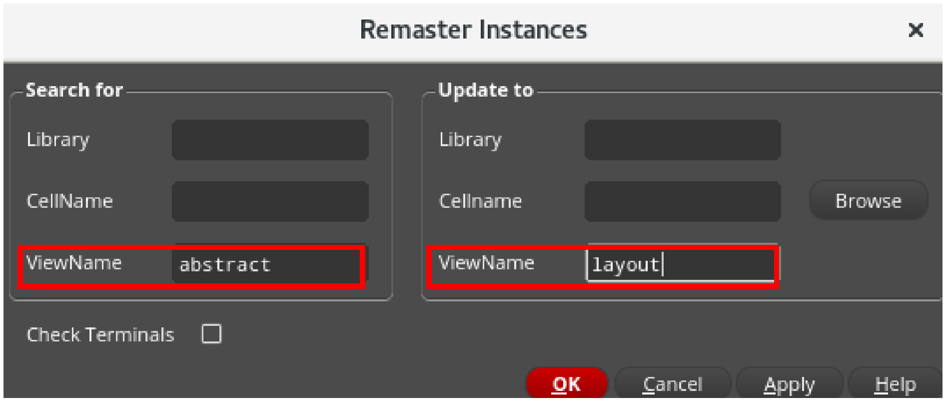
5. Enter the **CellName** *top_common* and the **ViewName** *abstract* to change just one instance to the *layout*.



6. Click **Apply**.

   The *top_common/abstract* view is replaced with the *layout* view.

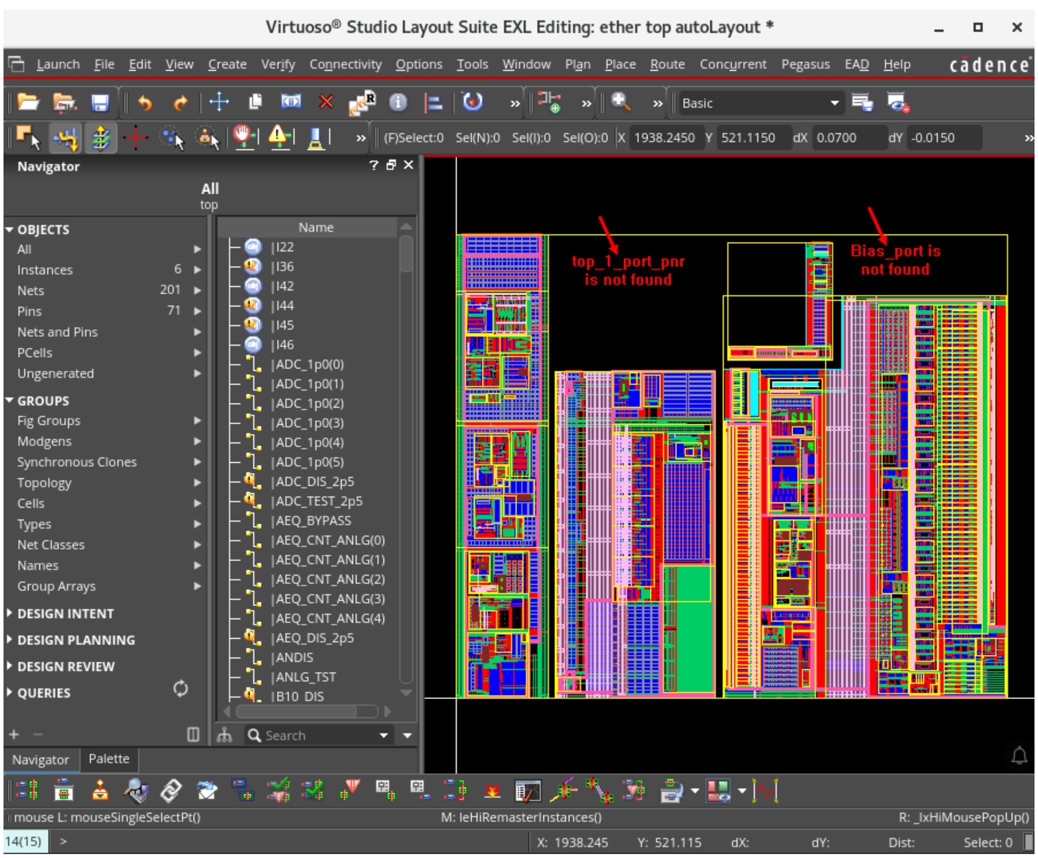
7. In the Remaster Instances form, remove *top_common* and leave *abstract* in the *Search for ViewName* field and *layout* in the *Update to ViewName* field, as shown here.



Note: By filling in only these values in the form, you update any abstracts in this cellview to use the *layout* view of the same *CellName* and *Library*.

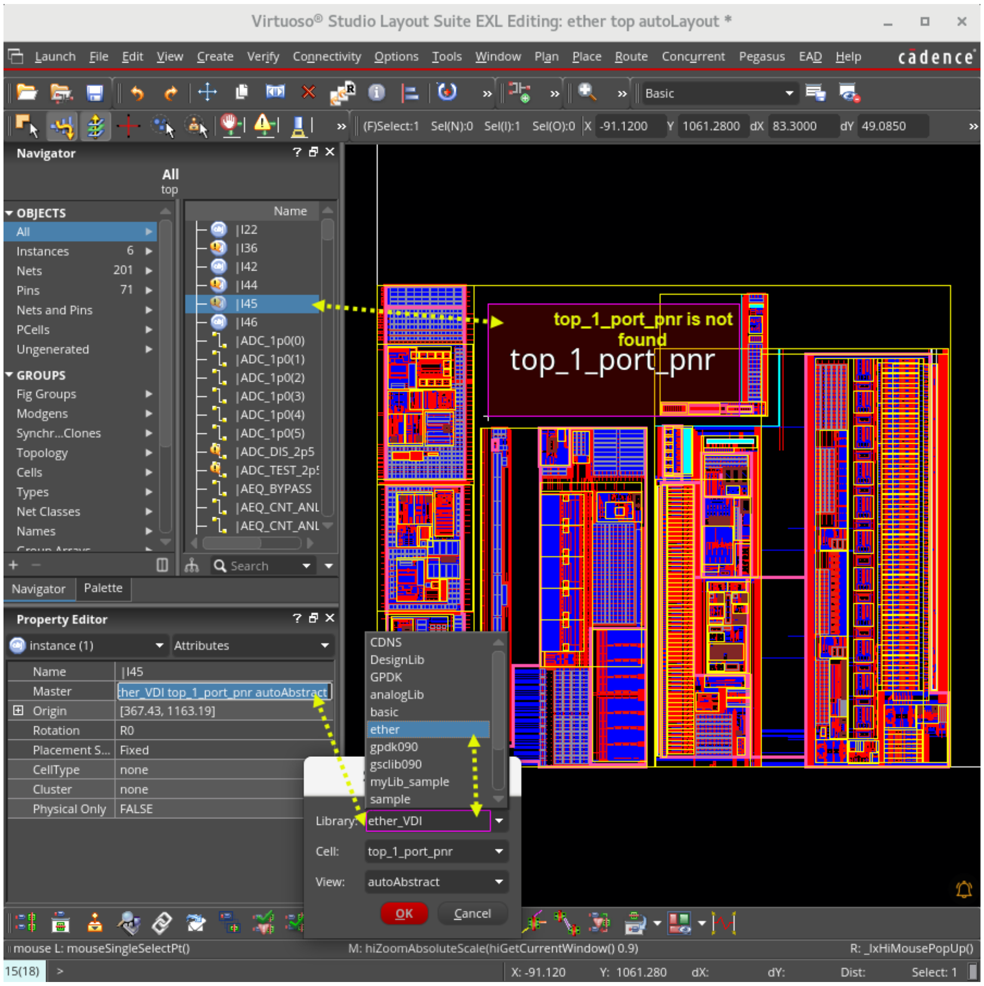
8. Click **OK** in the Remaster Instances form.

9. The flashing cellview instances *top_1_port_pnr* and *Bias_port* indicate that the cellviews have not been found in any of the available libraries.



10. Select the flashing instance *top_1_port_pnr* in the Property Editor assistant.

    In the Select **Master** form, which pops up:



a. Change the **Master** of the **Library** from *ether_VDI* to *ether*.

b. Change the **View** from *autoAbstract* to *layout*.

c. Click **OK**.

Note: Alternatively, you can use the **Edit – Basic – Properties** command or bindkey **Q** to query and change the properties.

The canvas updates to include the *top_1_port_pnr layout* view from the *ether* library.

11. Select the flashing instance ***Bias_port*** in the Property Editor assistant.

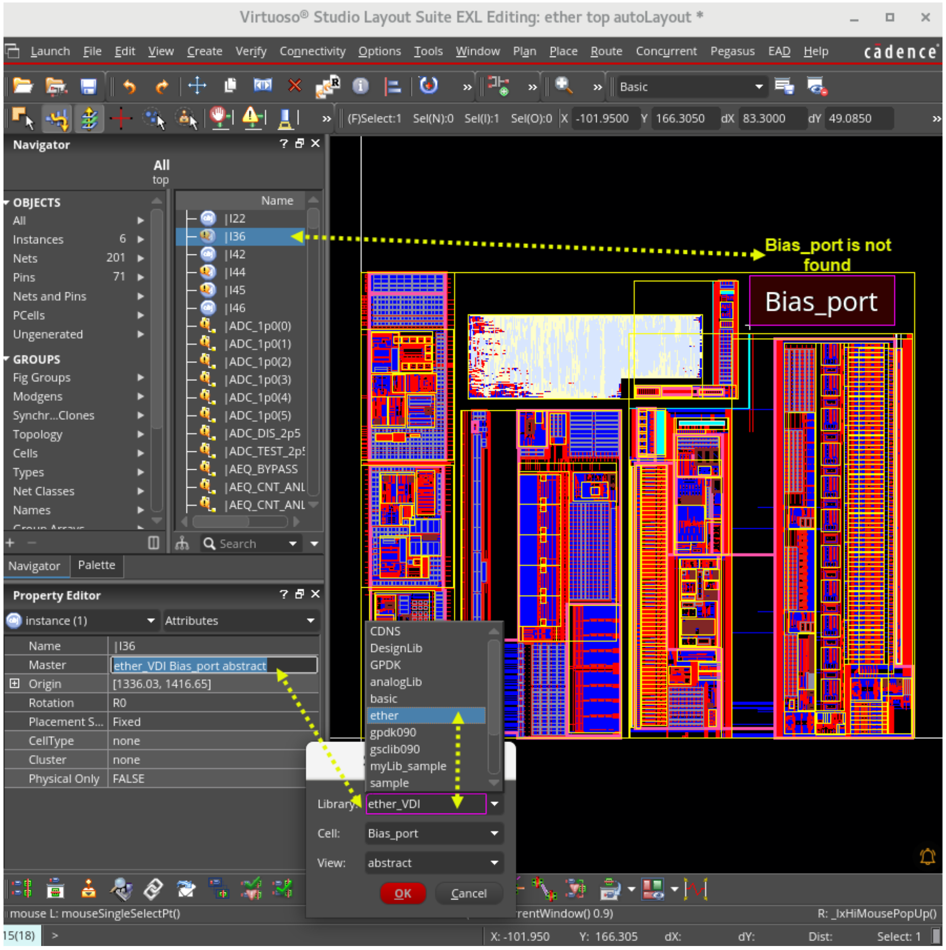
    In the **Select Master** form, which pops up:

a. Change the **Master** of the **Library** from *ether_VDI* to *ether*.

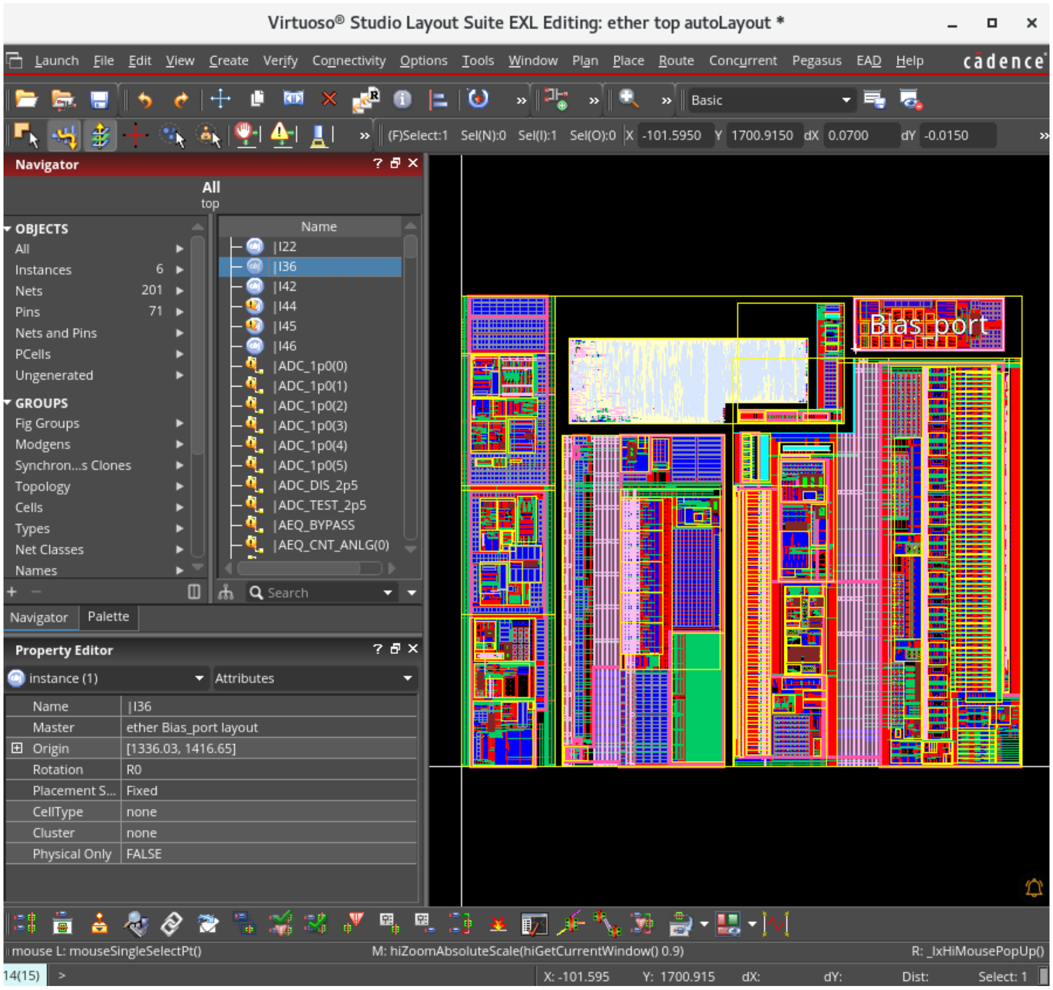
b. Change the **View** from *abstract* to *layout*.

c. Click **OK**.

The canvas updates to include the *Bias_port layout* view from the *ether* library.



Now all the cellviews have been changed to layout views, as shown here.

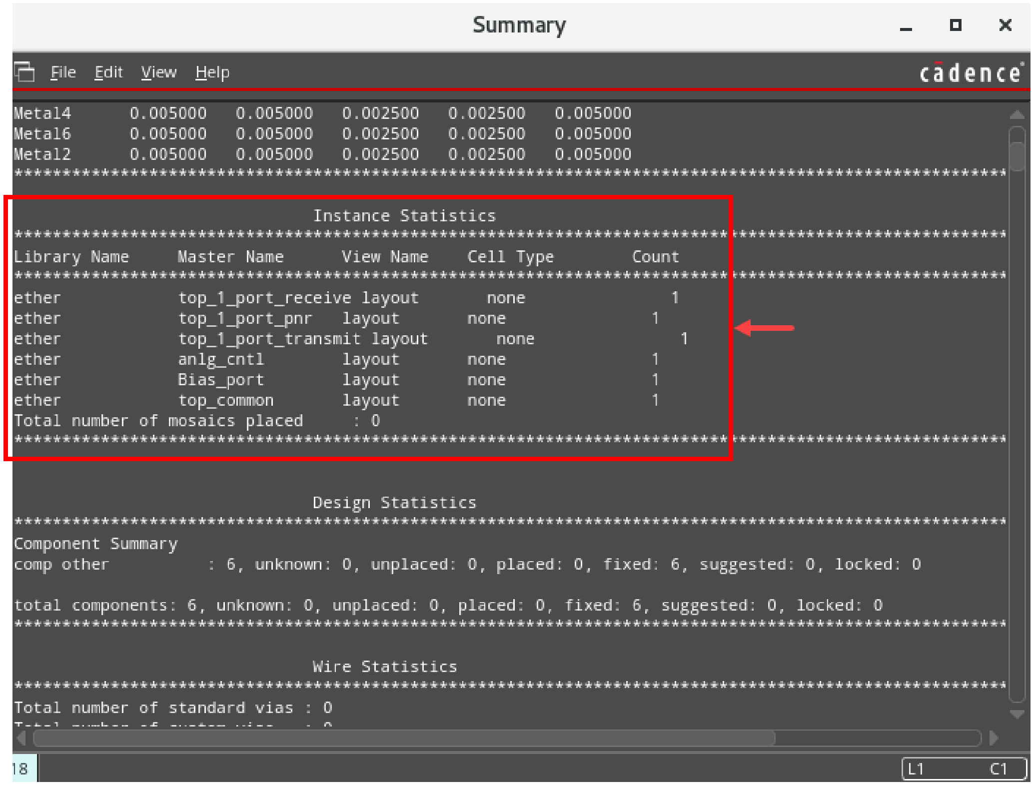


12. In the design canvas, choose **File – Summary** to open the Design Summary form.

    a. In the Design Summary form, click the **Select All** button.

    b. Click **OK** to open the *Summary* window.

Review the instances in the *Summary* to see them all show layout views, as shown here.



13. Choose **File – Close Window** to close the *Summary* window.

14. Close any open cellviews or forms. Do **NOT** save.

15. In CIW, choose **File – Exit** to close the Virtuoso platform.

16. Leave the Virtuoso session.

17. In the shell, enter the following command:

```
cd ~
```

This command places you in your home directory.

Summary

In this lab, you learned how to

- Use the *Remaster Instances* command to update the layout

End of Lab

Appendix A: Basic Commands

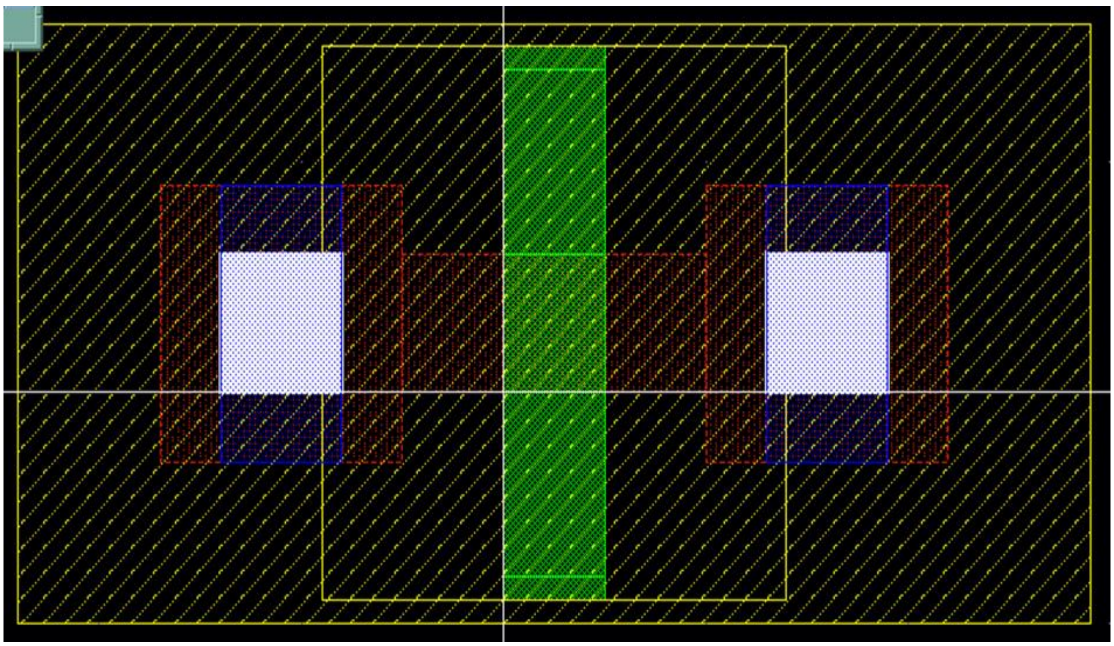
Lab A-1 Using Basic Commands (Optional)

Objective: To draw shapes that layout the NMOS and PMOS cellviews.

This optional lab shows you how to use the basic *Create* commands to build primitive cells.

In this lab, you build a basic *nmos* cell from scratch, then define it as a cell for further use.

Design Rules: nmos



- Oxide overlap of contact: **0.06u**
- Nimp overlap of all: **0.02u**
- Contact minimum width: **0.12u**
- Contact spacing: **0.12u**
- Contact to gate spacing: **0.16u**
- Poly extension: **0.16u**
- Metal overlap of contact: **0.02u**
- Min w/l: **.16/.16u**

Lab Setup

1. Follow the steps below to set up the environment and perform the lab:

2. If you continue from the previous lab, move to the sub-heading: Creating the *nmos* Cellview Otherwise, proceed to the next step.

3. Update the following line in the /VLPT2_IC_25_1/. cshrc_VLPT2 file with the appropriate valid local tool paths for Virtuoso®.

   ```
   setenv CDSHOME /ccstools/cdsind1/Software/IC251Base_lnx86
   ```

4. Save and close the file.

5. Source the updated class setup file.

   ```
   Source .cshrc_VLPT2
   ```

Creating the *nmos* Cellview

1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

   ```
   cd./APDA_VLPT2_BasicCommands
   ```

2. If the Virtuoso® session is not already open, invoke the Virtuoso executable by entering this command in the terminal window:

   ```
   virtuoso &
   ```

3. In the CIW, choose **File – New – CellView**, then enter these values in the form.

Library	`ether`
Cell	`nmos`
View	`layout`

4. Click **OK**. The cellview is empty.

   Referring to the design rules, continue with the following steps to draw the *nmos* cellview. The device size is:

   Width: **1.09**

   Length: **0.16**

5. In the layout canvas, choose the **Options – Display** menu commands or press the bindkey **E** to open the Display Options form.

6. Set the X and Y Snap Spacing to **0.01** each.

7. Set Minor Spacing to **0.01** and Major Spacing to **0.05.** Click **OK**.

8. Make the **Cont drawing** layer as the current layer.

9. In the design window, choose **Create – Shape – Rectangle** or press the bindkey **R** to invoke the Create Rectangle command.

10. Make one contact opening which is **0.16 x 0.16**.

    Enter **0:0** in response to the first corner and **0.16:0.16** in the input pane as the second corner for the first rectangle.

    You can also start the rectangle manually and watch the *dX* and *dY* coordinates to see the size of the rectangle.

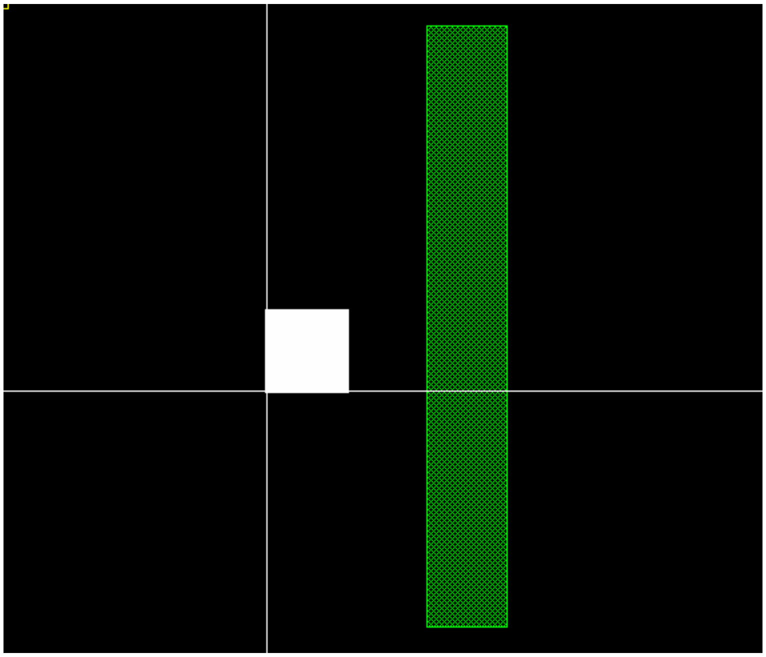
11. In the design window, in the Layers assistant, select **Poly drawing**.

12. Zoom in so that you can see the rectangle you created.

13. In the layout, choose **Tools – Create Measurement** or press the bindkey **K** to invoke the Create Measurement command.

    Use the Create Measurement command to measure distance.

14. Create a **Poly** rectangle with **0.16** spacing to contact and a width of **0.16**.



15. Select the **Cont** rectangle.

16. Choose **Edit – Copy** or press the bindkey **C** to invoke the Copy command.

17. Click on the **lower**-left of the **Cont** shape as the reference point.

18. Move the cursor **0.16** microns beyond the **Poly** shape and click to place the copy.

    Notice the *dX* and *dY* counters in the lower right of the viewport. You can use them as counters for distance and spacing if you start at a known location.

19. Press **Esc** to stop the copy command if it is open.

20. Select both contacts by using **Shift** and the left mouse button.

21. Choose **Edit – Copy** and press **F3** to bring up the Copy form.

22. Set *Rows* to **3** and leave *Columns* at **1**. You create an array of contacts vertically.

23. Click to set the reference point at the **lower**-left corner of the left contact.

24. Move the cursor up, watching the *dY* counter until it shows dY: **0.32**. This is the contact width and space. Click to place the lower row of the contact array.

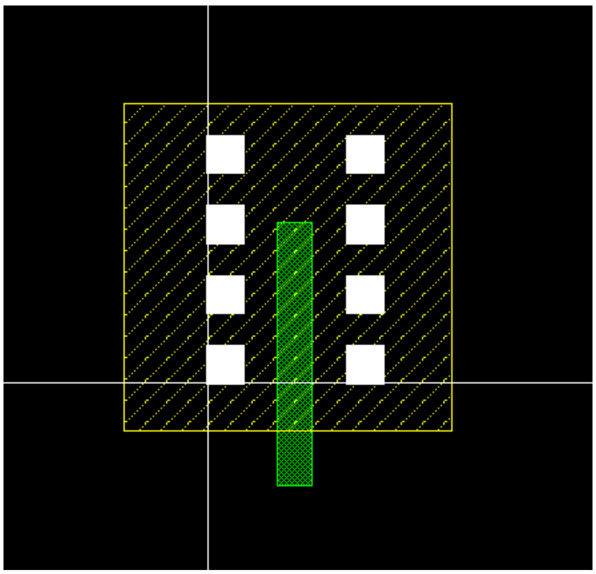
25. Continue up and notice the rubberbanding of the remaining contact array to be placed. Click when the *dY* counter indicates dY: **0.32** again. The remaining contacts are placed in the array.

26. Press **F** to fit the screen around the shapes.

27. In the design window, in the **Layers** assistant, select **Nimp drawing**.

28. Choose **Create – Shape – Rectangle** to invoke the Create Rectangle command.

29. Enclose the contacts with the **Nimp** layer. Do not worry about the spacing right now.

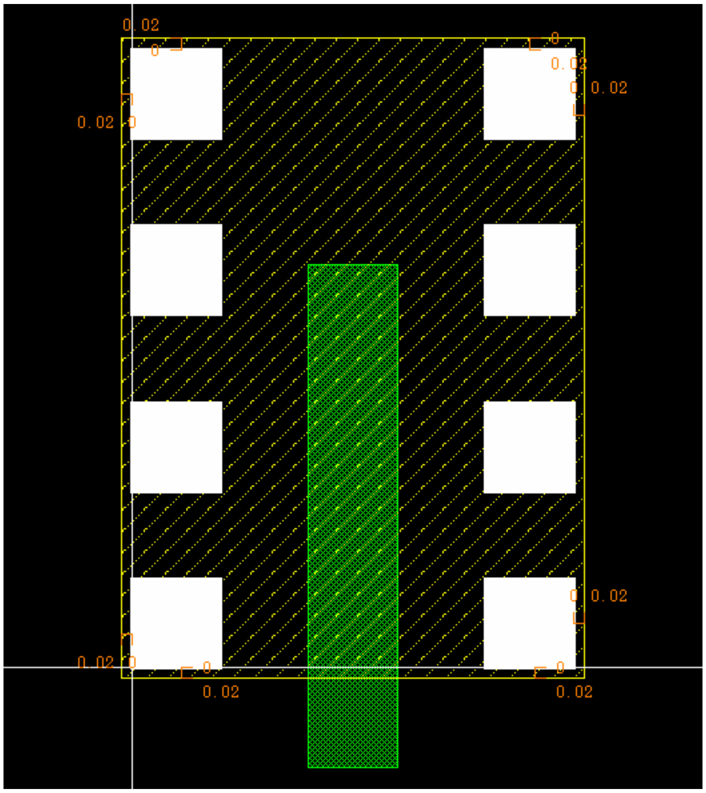


30. In the layout canvas, choose the **Edit – Stretch** menu command or use the **Stretch** icon in the Edit toolbar or press the bindkey **S** to invoke the Stretch command.

    Notice the **(P)Select** in the upper-middle indicating partial select. You can now stretch the edges or corners of the shapes. Click **F3** to open the form.

31. Press the bindkey **K** to start the Create Measurement command.

32. Click the **lower**-left corner of the **lower**-left contact and measure down to establish the ruler for the *Nimp* overlap of **0.02**. Click to finish the ruler. Create four rulers to show the measurements for each opposing corner, as shown here.



33. Press the **Esc** key to stop the Create Measurement command. The Stretch command is still active.

34. Select the **lower**-left corner of the **Nimp** layer and stretch it to the **0.02** overlap distance indicated by the rulers.

35. Press **F3** to set the **Snap Mode** to *anyAngle* to move both edges at once.

36. Select the **upper**-right corner also and stretch the edges to **0.02** overlap.

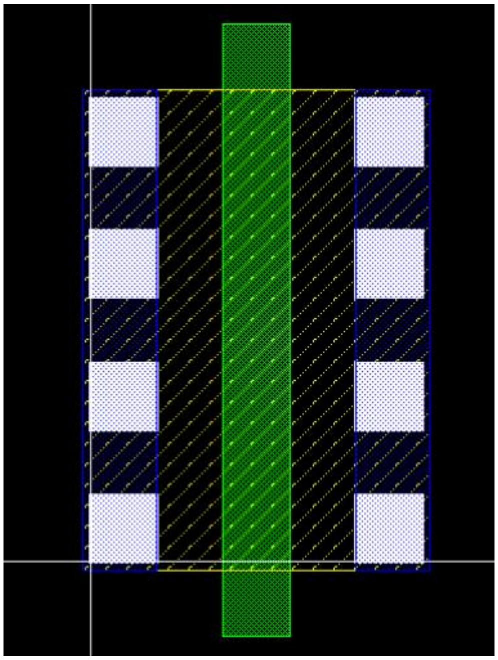
    It is best to set the ruler to *orthogonal* for polar measurements.

37. Stretch the **Poly** shape to extend beyond the **Nimp** by **0.16**.

38. Press the bindkey **Shift+K** to remove the rulers.

39. In the design window, in the Layers assistant, select **Metal1 drawing**.

40. Create **Metal1** rectangles around the contacts with **0.02** enclosure, as shown here.

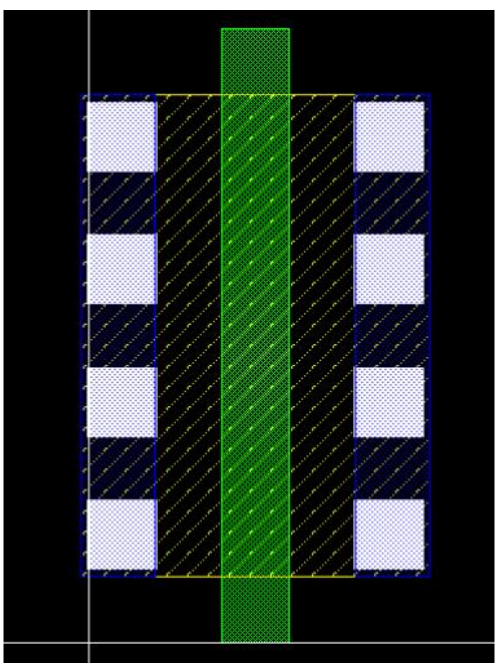


41. Choose **File – Save** in the viewport to store the cellview.

42. In the design window, choose **Edit – Advanced – Move Origin**.

    The cursor becomes a complete screen axis, so you can align the left and bottom edges of the design to establish a new **0:0** coordinate, which is coincident with the **lower**-left data extremes.

    You are prompted to point at the new location of the origin.



43. Click to establish the new origin.

44. Choose **File – Save** to save the design.

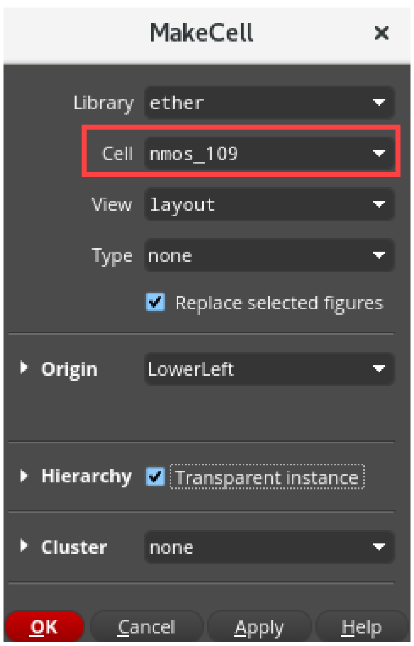
45. Choose **View – Zoom To Fit All** (bindkey **F**) to fit the design window.

Using the Flat Data to Make a New Cell

1. Select all the data in the cellview by clicking and dragging around all the shapes or by choosing **Edit – Select – Select All** or by pressing the bindkey **Ctrl+A**.

2. In the layout window, choose **Edit – Hierarchy – Make Cell.**

   In the MakeCell form, enter the values, as shown here.



3. Click **OK**.

   The *nmos_109* cellview is added to the *ether* library, and the shapes are replaced by the cell instance.

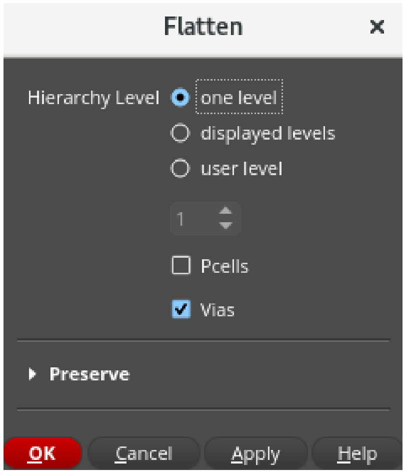
Flattening a Cellview

1. Make sure that in the Objects assistant, **Instances** are selectable in the layout.

2. Select the **nmos_109** instance in the layout canvas.

3. Press **Shift+F** to view the hierarchy of the layout design.

4. In the layout, choose **Edit – Hierarchy – Flatten** to open the Flatten form.



5. Click **OK** with the default values as shown here.

   The shapes are returned to level **zero** data.

   Note: The *ether/nmos_sample & ether/nmos_109_sample* cells are available for reference.

6. In the layout canvas, choose **File – Save**.

7. Close any open forms and cellviews.

8. Leave the Virtuoso session.

9. In the shell, enter the following command:

   ```
   cd ~
   ```

   This command places you in your home directory.

Summary

In this lab, you learned the following commands:

- Create Rectangle
- Copy
- Create Measurement
- Stretch
- Move Origin
- Make Cell
- Flatten

End of Lab

Appendix B: Polygons and Circles

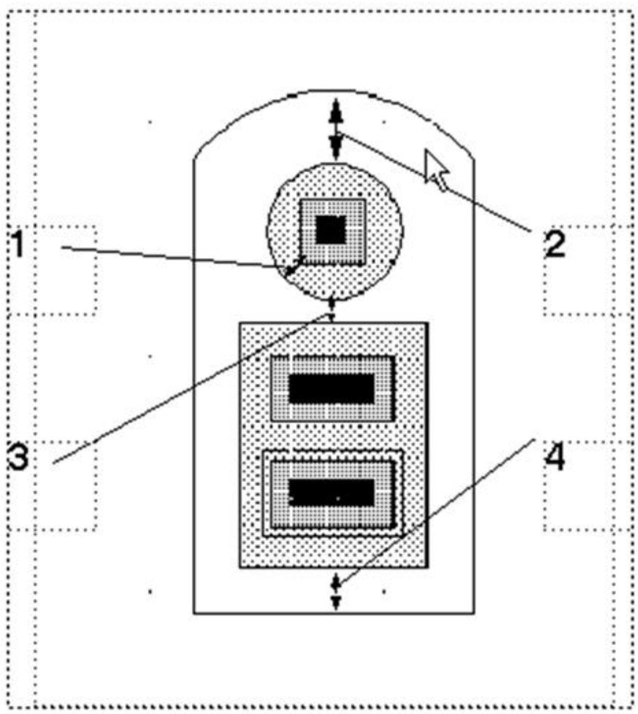
Lab B-1 Creating Polygons and Circles (Optional)

Objective: To draw the collector and buried layer for the npn device with the Create Circle and Create Polygon commands.

In this optional lab, you use the Create Circle and Create Polygon commands.

Use the following design rules to create the correct device size and the correct spatial relationships between layers of the *npn* device.

Design Rules: npn



- ndiff overlap of contact: **1.0u**
- buried overlap of ndiff: **1.5u**
- ndiff to pbase spacing: **0.5u**
- buried overlap of pbase: **1.0u**

Lab Setup

1. Follow the steps below to set up the environment and perform the lab:

2. If you continue from the previous lab, move to the sub-heading: Drawing the *npn* Cellview Otherwise, proceed to the next step.

3. Update the following line in the /VLPT2_IC_25_1/. cshrc_VLPT2 file with the appropriate valid local tool paths for Virtuoso®.

   ```
   setenv CDSHOME /ccstools/cdsind1/Software/IC251Base_lnx86
   ```

4. Save and close the file.

5. Source the updated class setup file.

   ```
   Source .cshrc_VLPT2
   ```

Drawing the *npn* Cellview

1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

   ```
   cd ./APDB_VLPT2_PolygonsAndCircle
   ```

2. In the CIW, choose **File – Open** and enter the following values in the form:

3. In the CIW, choose **File – New – Cellview**, and enter these values in the form.

Library	ether
Cell	npn
View	layout

4. Click **OK**. The cellview is empty.

Referring to the design rules, continue with the following steps to create the *npn* cellview.

Using the Create Circle Command to Create the Collector

You use the **Create Circle** command to create the collector of the *npn*.

1. In the layout, change the current drawing layer to **Nimp drawing**.

   Note: You can use the Search Filter to find the required **LPP**.

2. Choose **Options – Display** and set the X and Y Snap Spacing to **0.1**.

3. Choose **Create – Shape – Circle** in the design window to invoke the Create Circle command.

4. Click for the center placement of the circle, X: **3.0** and Y: **8.0**. You can alternatively enter **3:8** in the input pane of the CIW.

5. Look at the *dX* and *dY* coordinates of your cursor. Move your cursor in the X direction **1.5u**.

6. Click to finish the circle. The diameter of the circle should be **3.0u**.

7. Press the **Esc** key to stop the Create Circle command.

8. In the design window, choose **Options – Editor** or press the bindkey **Shift+E** to open the Layout Editor Options form.

9. Change *Conic Sides* to ***200***. Click **OK**.

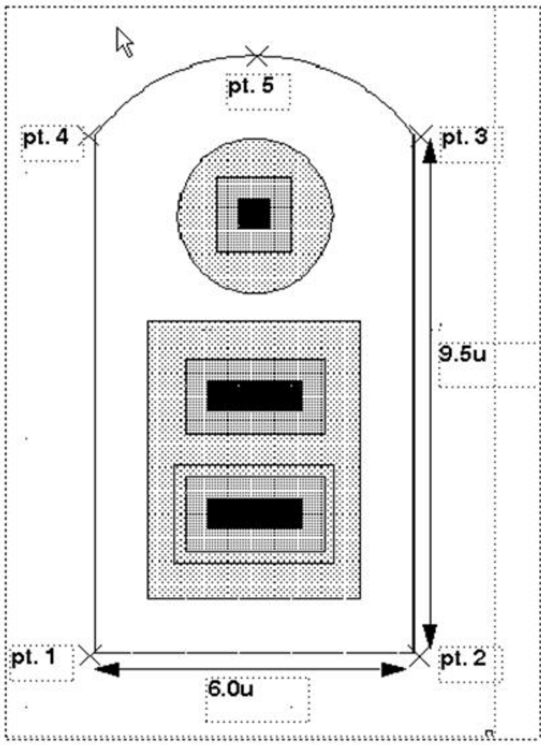
   After you convert the circle to a polygon, it will have **201** vertices.

10. To change the conic shape to a polygon, in the layout, select the circle and choose **Edit – Convert – To Polygon**.

    The circle has become a polygon with **201** vertices to complete the **200-sided circle**.

Using the Create Polygon Command to Draw the Nburied Layer

You use the **Create Polygon** command to draw the **Nburied** layer of the *npn*.



1. In the layout, choose **Create – Shape – Polygon** or press the bindkey **Shift+P** to invoke the Create Polygon command.

2. Change the current layer to **Nburied drawing**.

3. To start the polygon, click the first point in the **lower**-left corner **0:0** coordinate.

4. Buried layer overlap of **pbase** layer is **1.0u**.

5. Click to enter the second point, **6.0u**, in the X direction.

   As you enter each point, a dotted line shows you how the polygon closes if you stop at that point.

6. Move **9.5u** in the Y direction. Click to enter the third point.

7. Press **F3** to open the Create Polygon form.

8. Move your cursor to the **Create Polygon** form. Click **Add an arc segment**.

9. Move **6.0u** left in the X direction (**0:9.5** coordinate) and click to enter the fourth point.

   The cursor might tell you that this angle is not allowed. You can ignore this message for the moment. It is the result of the location of the cursor, which would create an illegal arc segment.

10. Center your cursor over the *ndiff* circle (**3.0** in the X direction). Move in the Y direction **1.5u** (dY: **1.5**). Click the fifth point to create the arc.

11. Press **Return** or **double**-click to finish the last leg of the polygon.

12. Press the **Esc** key to stop the Create Polygon command.

13. Select the ***Nimp*** polygon in the layout window.

14. Choose **Edit – Copy** or press the bindkey **C** to invoke the Copy command.

15. Click **F3 to open the Copy form.** Set *Array Rows* to **1**.

    You place the copy of the *Nimp* polygon right over the original polygon.

16. In the CIW, enter **0:0** for the reference point.

17. Enter **0:0** for the destination point.

    You copied the *Nimp* polygon over itself using **0:0** as a reference and destination points.

18. Press **Esc** to end the Copy command if it is still open.

    You are going to change one of the polygon circles into *Oxide*.

19. In the layout, press the bindkey **Q** to open the Edit Polygon Properties form.

20. Select the **Attribute** tab.

21. Open the ***Layer*** *cyclic* field by clicking the drop-down menu.

22. Change the layer name to **Oxide drawing**.

23. Click **Apply**.

24. Click in the circle and see whether the Properties form changes from *Oxide* to *Nimp*. You can cycle between layers by clicking where they overlap.

    Leave the *Nimp* shape selected.

25. In the design window, choose **Edit – Advanced – Size**.

    Because you selected **Nimp**, you oversize it by **0.1u**.

26. Enter **0.1** in the *Increase Size By Value* field. Click **OK**.

27. Click in the smaller circle. Check the Properties form to be sure you have **Nimp** selected.

28. Choose **Edit – Move** (bindkey **M**). Press **F3** to open the Move form.

29. Set the *Change To Layer* cyclic field to **Metal1 drawing**.

30. Leave Delta X Y values at **0**.

31. Click **Apply**.

    The *Nimp* shape moves to the *Metal1 drawing* layer.

32. In the Layers assistant, select **Cont drawing** in the design canvas.

33. Choose **Create – Shape – Rectangle** (bindkey **R**) to invoke the Create Rectangle command.

34. Add a **Cont** rectangle **1.6 x 1.6** in the emitter area.

    Watch the *dX* and *dY* values as you create the rectangle.

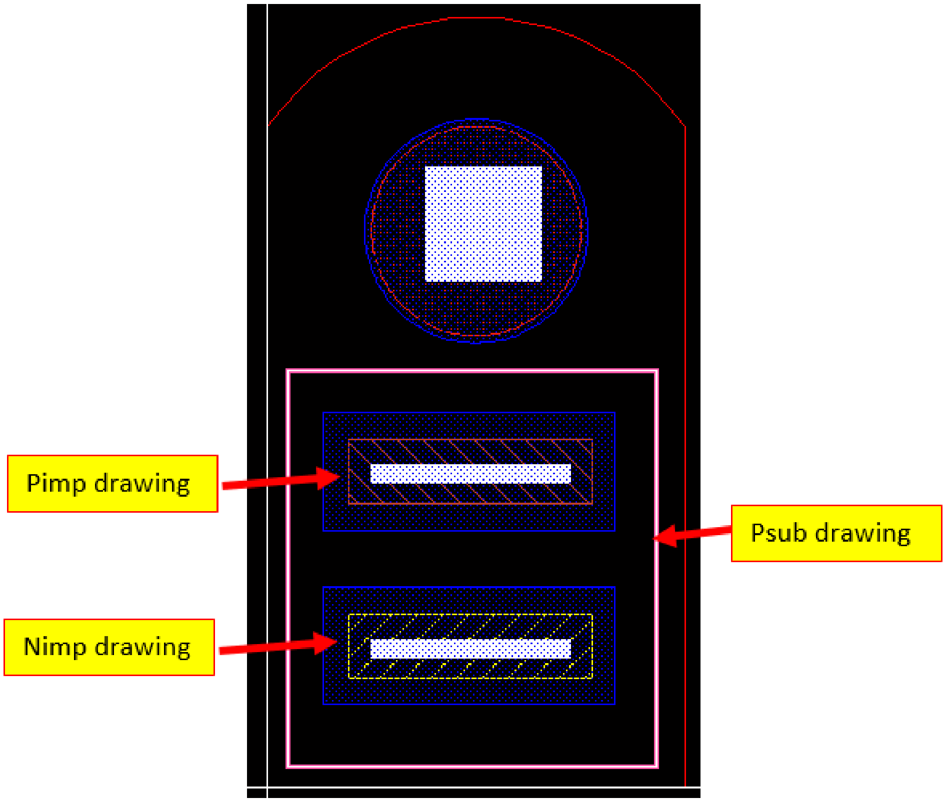
35. Create the **Psub** rectangle.

36. Add the **Pimp** rectangle with Metal1 and Contacts.

37. Add the **Nimp** rectangle with Metal1 and Contacts.

38. Choose **File – Save** to save the design and close the window.

39. In CIW, choose **File – Exit** to close the Virtuoso® platform.



Note: The *ether/npn_sample* cell is available for reference.

Summary

In this lab, you learned how to

- Create a circle
- Convert circle into a polygon
- Create a polygon
- Create an arc
- Change layers and create various shapes
- Copy
- Resize
- Change layers with the Move command

End of Lab